AthenaK shock encounters Implementation and test report

Recorded local evidence | 8 October 2026 | Branch: codex/shock-multiphase

What was implemented

On-the-fly shock detection and persistent per-particle shock-encounter histories. The live C++/Kokkos detector samples the completed fluid state; the tracker groups particle observations into episodes; migration, output and restart preserve those episodes.

Component	Purpose
Live mesh detector	Port the frozen ShockIt numerical contract and retain the unreduced shock region.
Encounter tracker	Record entry/exit brackets, positions, Mach estimates, normals and validity/censoring flags.
Persistence	Move complete typed state with each particle; use bounded event output and source-bound restart sidecars.

Current result

Local engineering checks pass. The simple Mach 2 pilot records 128 completed episodes for 128 particles in each of eight seeds. Strong and boosted fixtures expose additional episodes and strength bias. Human scientific approval and R1 release acceptance remain pending.

How to read this report

Pages 2-29 pair each original vector plot with its own test setup, reading guide, measured result and limitation directly above it. General methods, persistence details and reproducible run instructions follow in the appendix on pages 30-36.

The existing lagrangian_mc tracer makes stochastic parent-cell hops. Positions are cell centers. Lines in trajectory plots are sample sequences, and timing brackets describe neighboring shock-region observations. Neither is an exact continuous material crossing.

Evidence source: production commit 3de18d97; selected base multiphase_turb 526a187f938e38634e89a88192ebe9731c4d36bb. This report adds explanation to recorded results; no simulations or numerical thresholds were changed to produce it.

Test 01 | 2d dens

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. This page shows native density snapshots.

How to read the plot

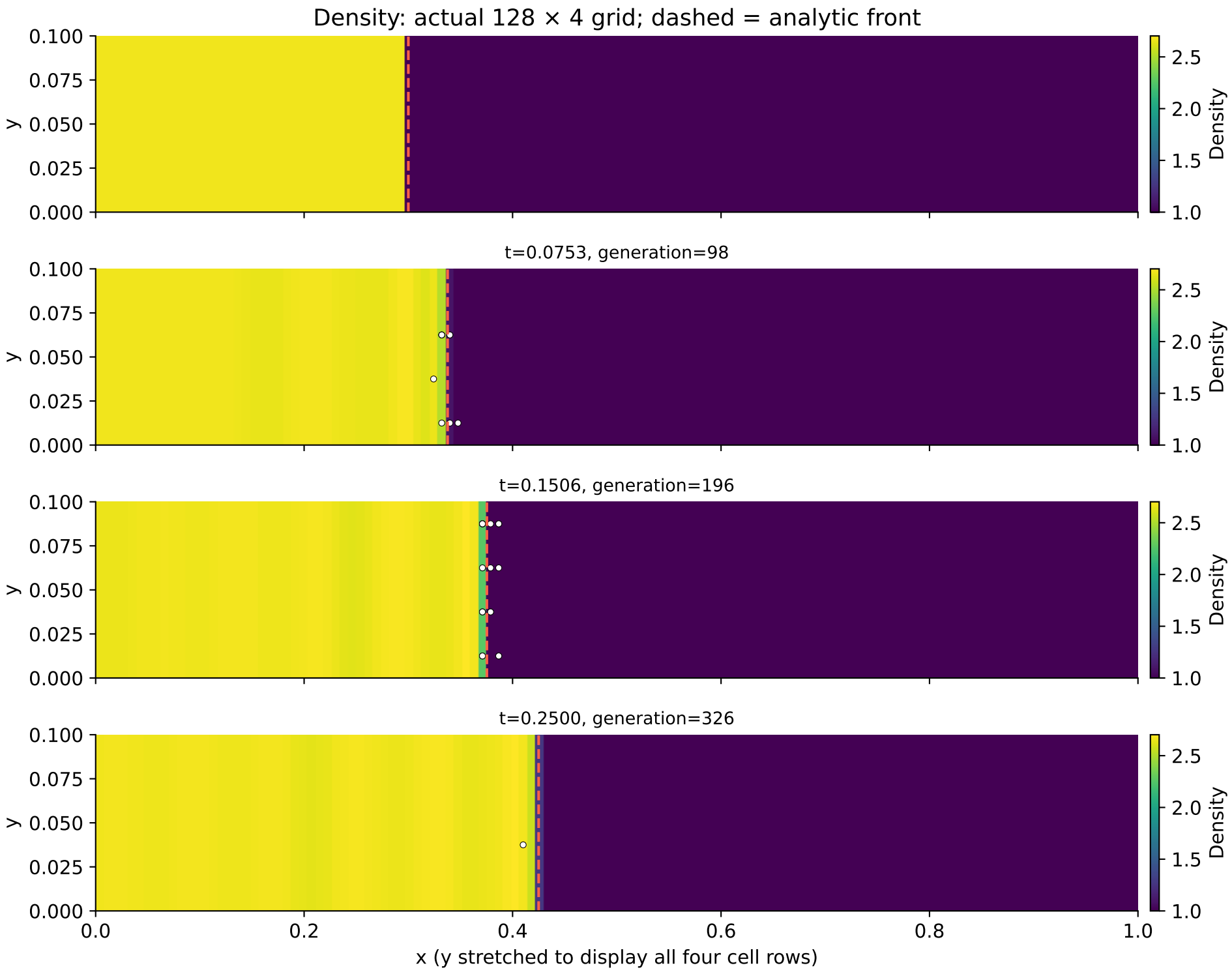
Read top to bottom from t=0 to t=0.25; the middle panels show their actual recorded times. Color is density. Dashed red marks the analytic front; white dots are sampled particles inside the unreduced shock region.

Measured result

The independent jump conditions prescribe density 1 upstream and 2.6667 downstream. The front moves right to x=0.425 at t=0.25. Synchronization checks separately match particle time, cell position and density exactly to native thermo observations.

What this establishes

The y direction is stretched to display four real grid rows. White dots are particle samples, not a full mesh mask. This planar fixture does not qualify curved, interacting or turbulent 2D shocks.



Test 02 | 2d pressure

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. This page shows native pressure snapshots.

How to read the plot

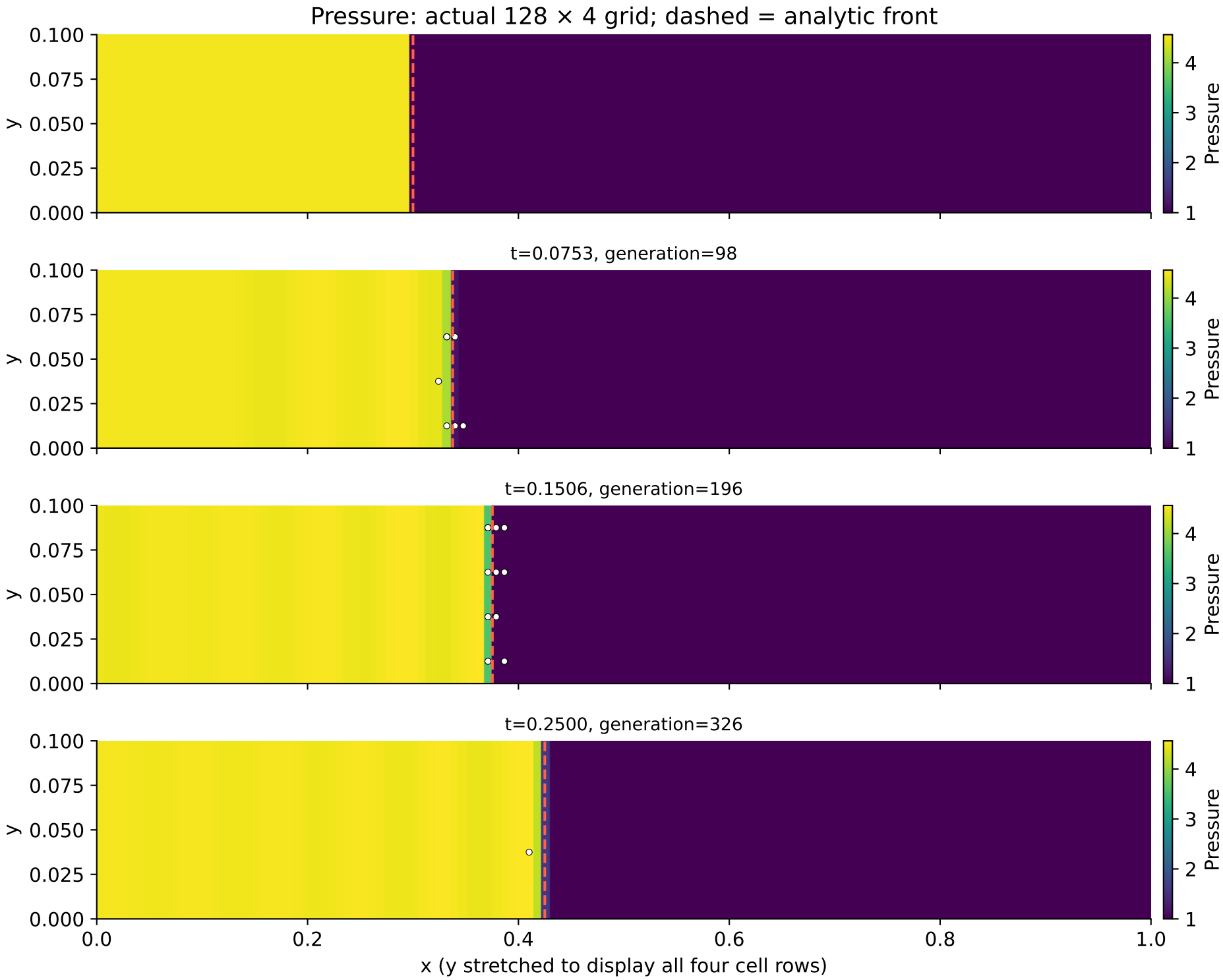
Read top to bottom from t=0 to t=0.25; the middle panels show their actual recorded times. Color is pressure. Dashed red marks the analytic front; white dots are sampled particles inside the unreduced shock region.

Measured result

The independent jump conditions prescribe pressure 1 upstream and 4.5 downstream. The front moves right to x=0.425 at t=0.25. Synchronization checks separately match particle time, cell position and density exactly to native thermo observations.

What this establishes

The y direction is stretched to display four real grid rows. White dots are particle samples, not a full mesh mask. This planar fixture does not qualify curved, interacting or turbulent 2D shocks.



Test 03 | 2d temperature

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. This page shows native temperature proxy P/rho snapshots.

How to read the plot

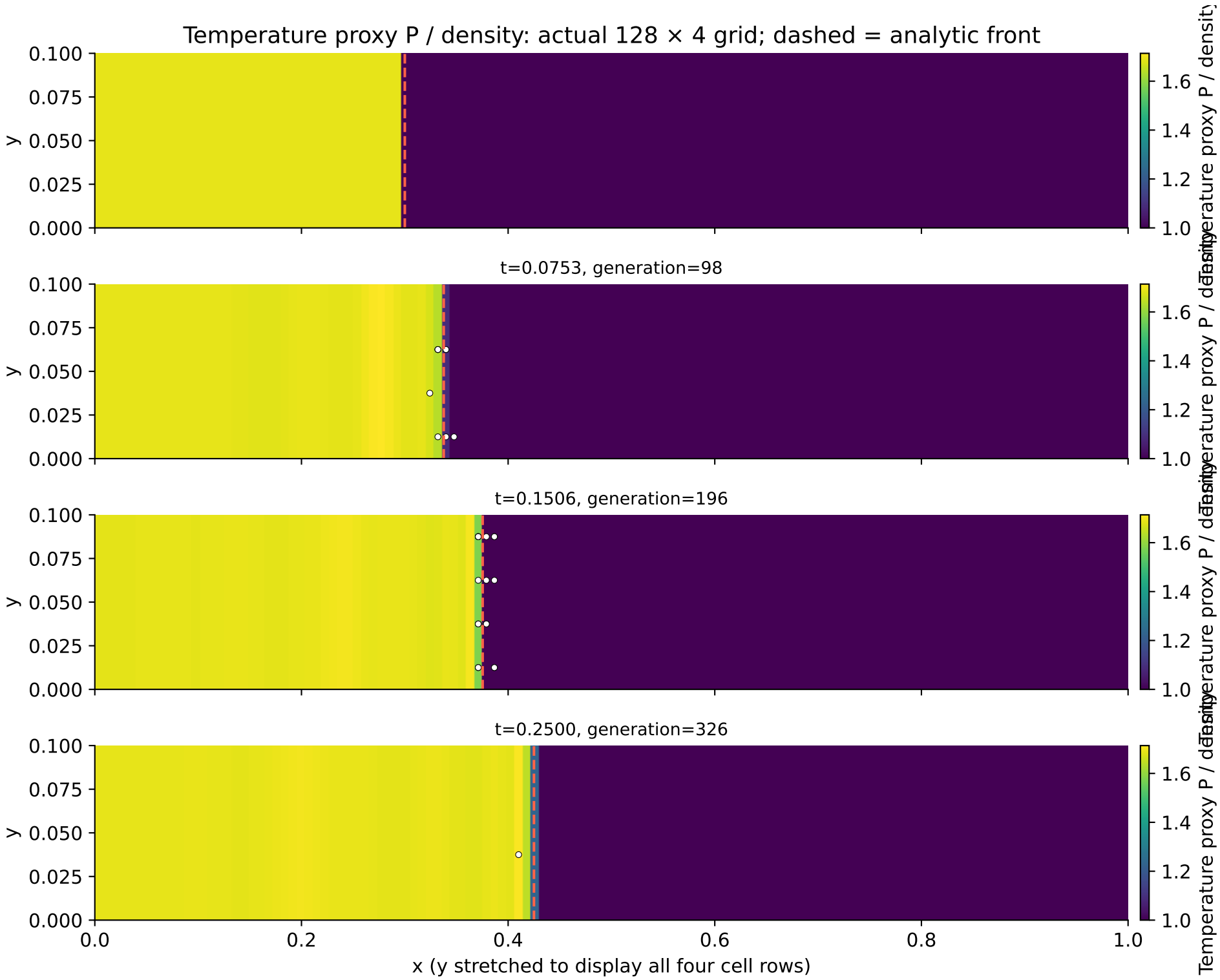
Read top to bottom from t=0 to t=0.25; the middle panels show their actual recorded times. Color is temperature proxy P/rho. Dashed red marks the analytic front; white dots are sampled particles inside the unreduced shock region.

Measured result

The independent jump conditions prescribe P/rho 1 upstream and 1.6875 downstream. The front moves right to x=0.425 at t=0.25. Synchronization checks separately match particle time, cell position and density exactly to native thermo observations.

What this establishes

The y direction is stretched to display four real grid rows. White dots are particle samples, not a full mesh mask. This planar fixture does not qualify curved, interacting or turbulent 2D shocks.



Test 04 | rh profiles

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. Initial states satisfy mass, momentum and energy conservation in the shock frame.

How to read the plot

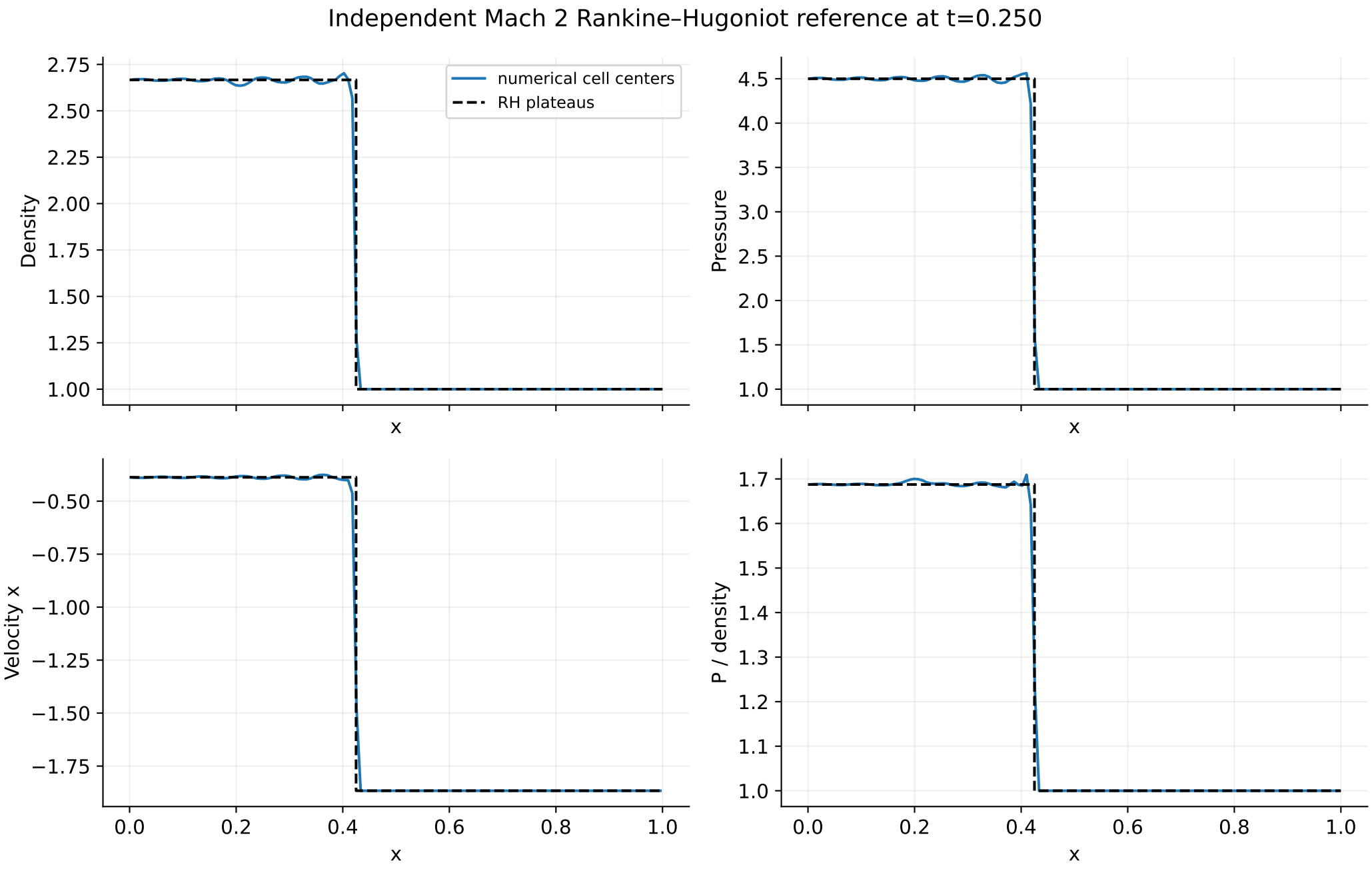
The four panels compare numerical density, pressure, x velocity and P/rho with dashed independent Rankine-Hugoniot plateau values at t=0.25.

Measured result

The reference front is x=0.425. Density ratio is 8/3 and pressure ratio is 4.5; velocities are -1.86643 upstream and -0.38741 downstream. The resolved profiles show finite shock thickness and small downstream oscillations.

What this establishes

This checks the fluid against a physical reference independent of ShockIt. It does not establish a plateau-certified particle Mach estimator or an exact particle crossing time.



Test 05 | shock space time

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. Native grid snapshots are written throughout the run.

How to read the plot

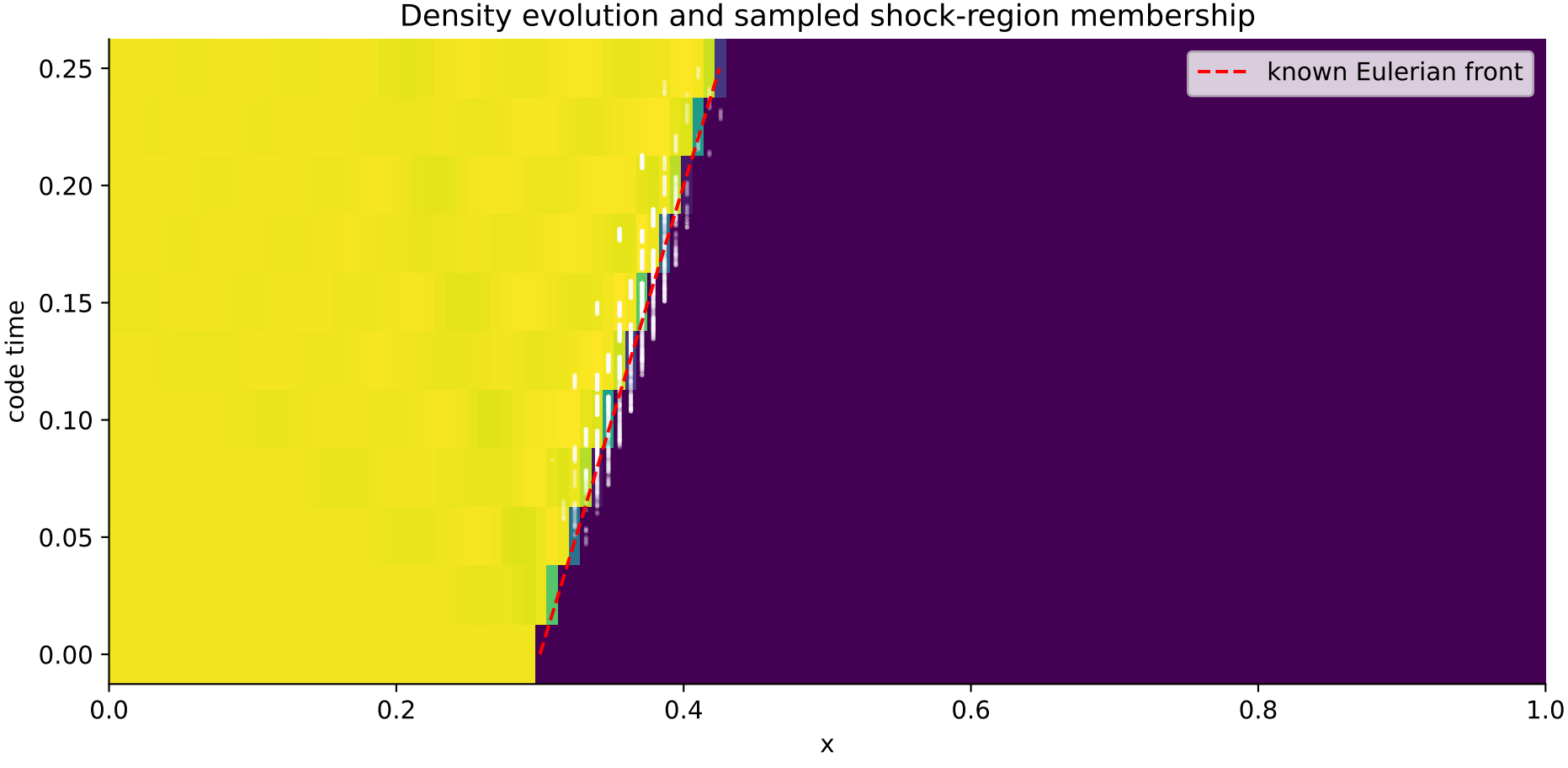
Horizontal coordinate is x; vertical coordinate is simulation time. Color shows density. The red dashed line is the prescribed Eulerian front; white dots mark synchronized inside-region particle observations.

Measured result

The density jump follows the translating front, and the sampled membership appears around the resolved shock. At the end the analytic front is x=0.425. The matching particle timestamp/position/density tests passed.

What this establishes

Space-time interpolation of sparse grid outputs is a visualization. It is not a continuous material trajectory or a fitted crossing-time oracle.



Test 06 | front position

Test setup

Mach 2 planar shock, gamma=1.4, RK2/PLM/HLLC, 128 x 4 cells, 128 MC tracers, CFL=0.3, t_limit=0.25. The known front is x=0.3+0.5t. The fluid front is measured by linearly interpolating its density midpoint.

How to read the plot

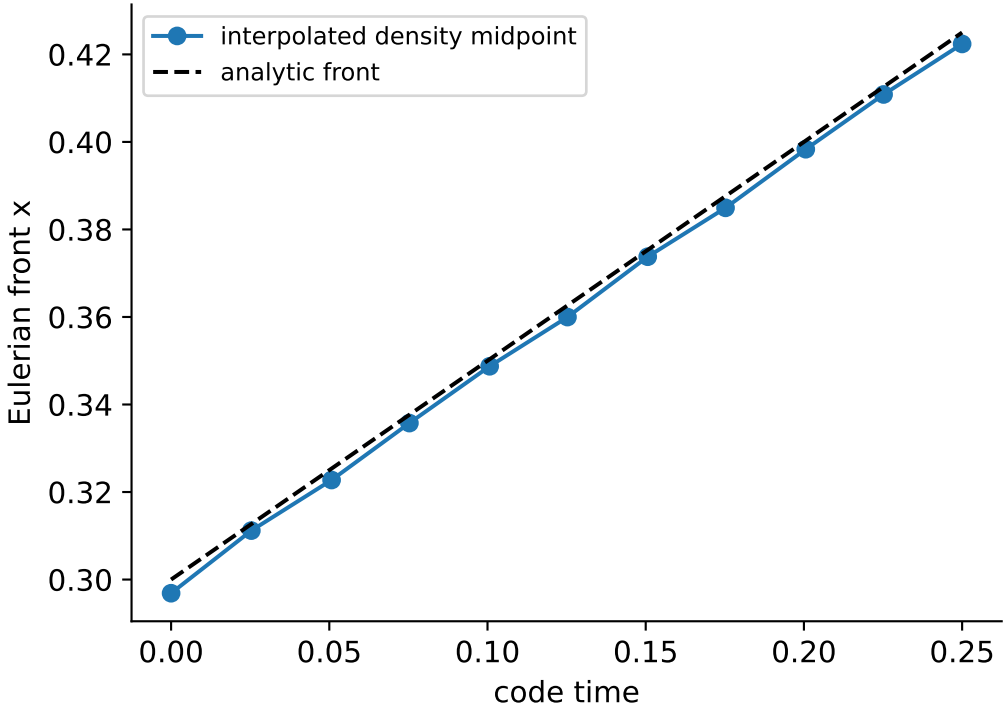
Left: midpoint location versus the known front. Right: midpoint minus analytic position, divided by the x cell width; zero means coincident locations.

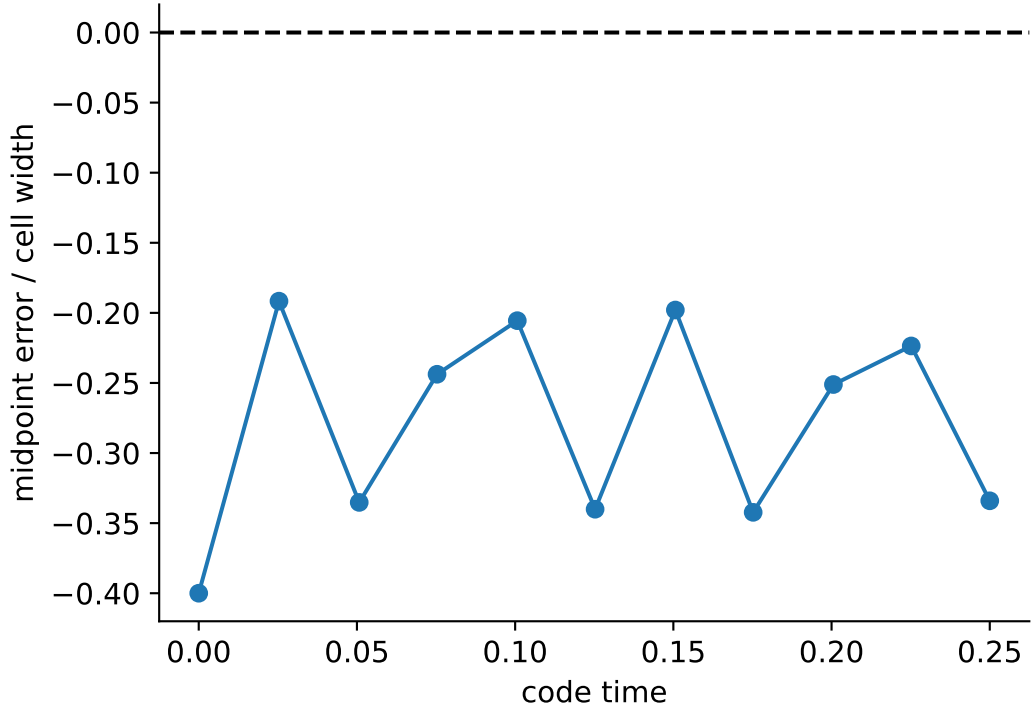
Measured result

The recorded midpoint offsets range from -0.400 to -0.192 cells across 11 snapshots. Initial cell placement contributes an offset; the midpoint is required to have a unique crossing.

What this establishes

This is an Eulerian front diagnostic, not a particle timing estimate. No physical acceptance tolerance is inferred from the displayed offset.





Test 07 | seed11 parents

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

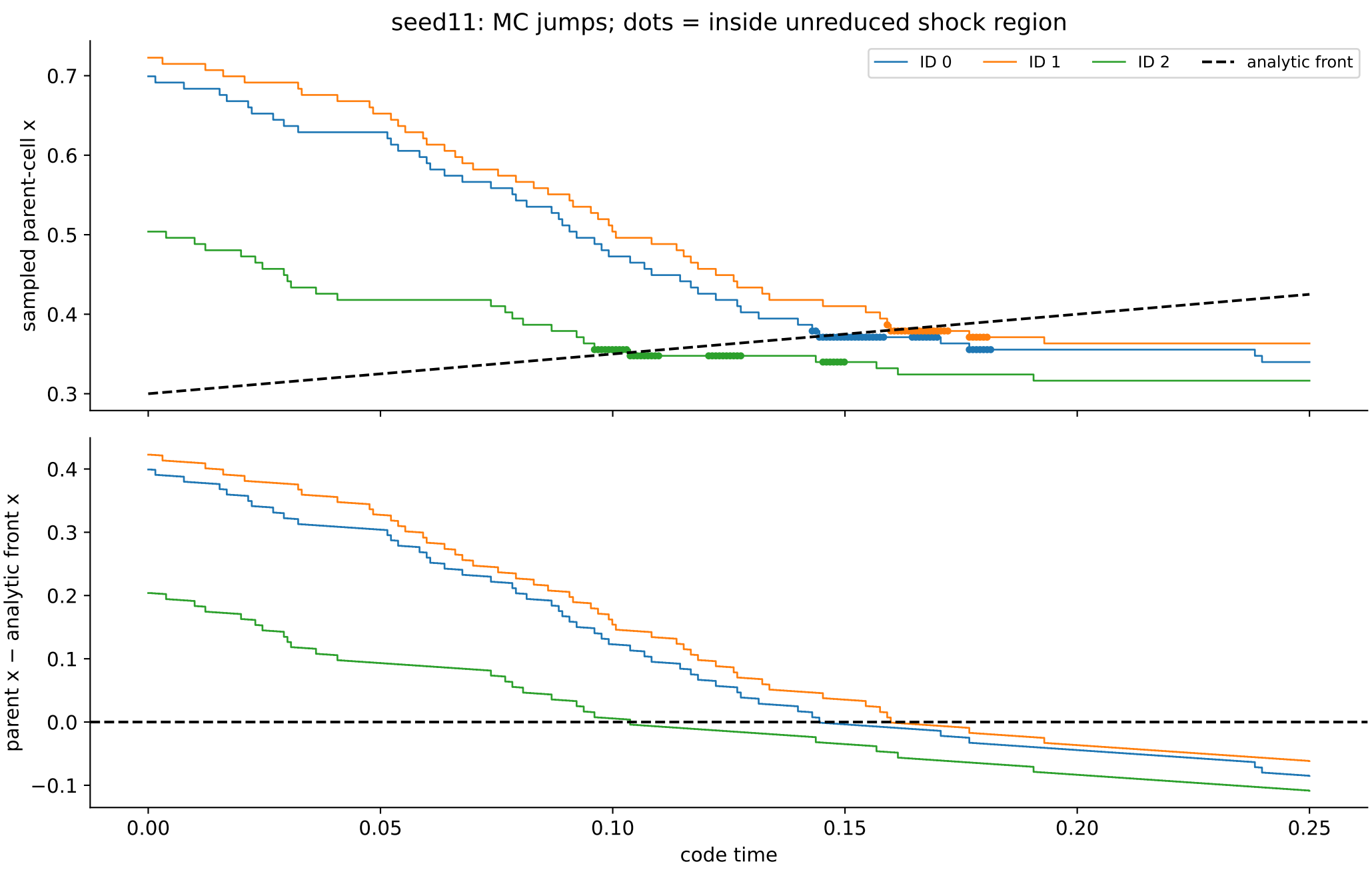
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0163 to +0.0118 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 08 | mach1.2 parents

Test setup

Input Mach 1.2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.45. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

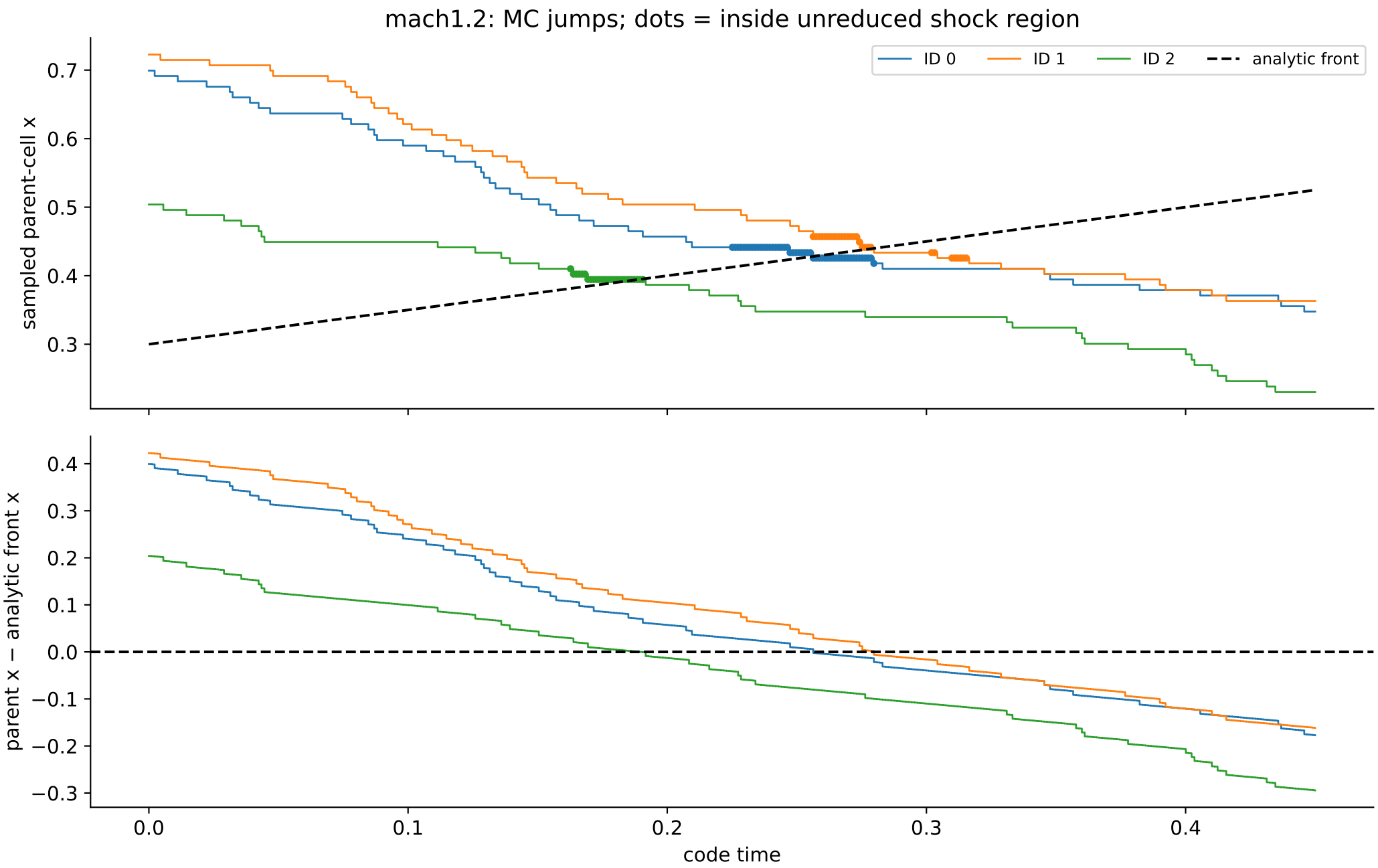
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.1031 to -0.0157 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 09 | mach1.5 parents

Test setup

Input Mach 1.5, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.3. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

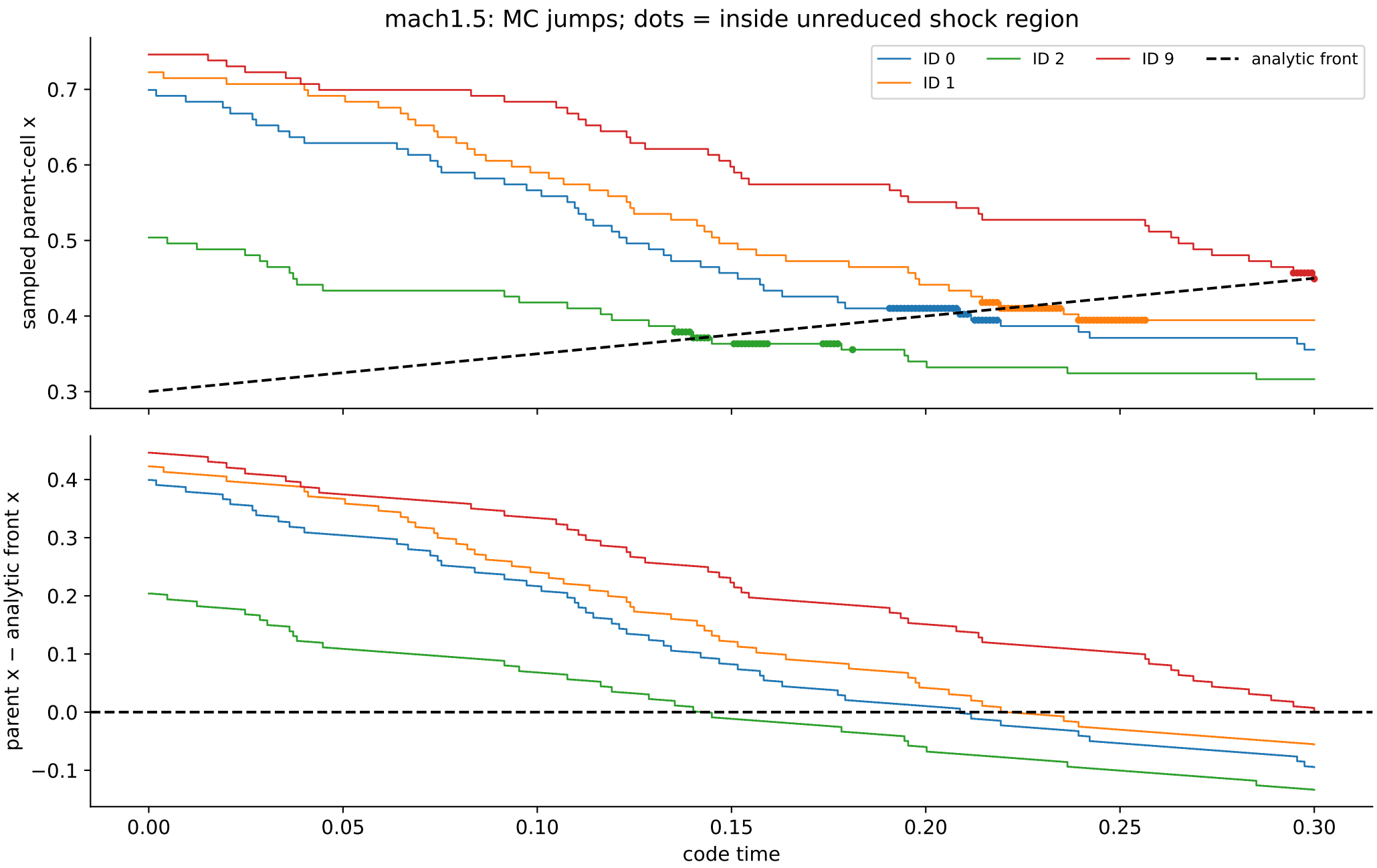
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

127 completed numerical episodes for 127 unique particles; 1 suspended. First-inside pressure-Mach error is -0.0093 to +0.0099 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 10 | mach3 parents

Test setup

Input Mach 3, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

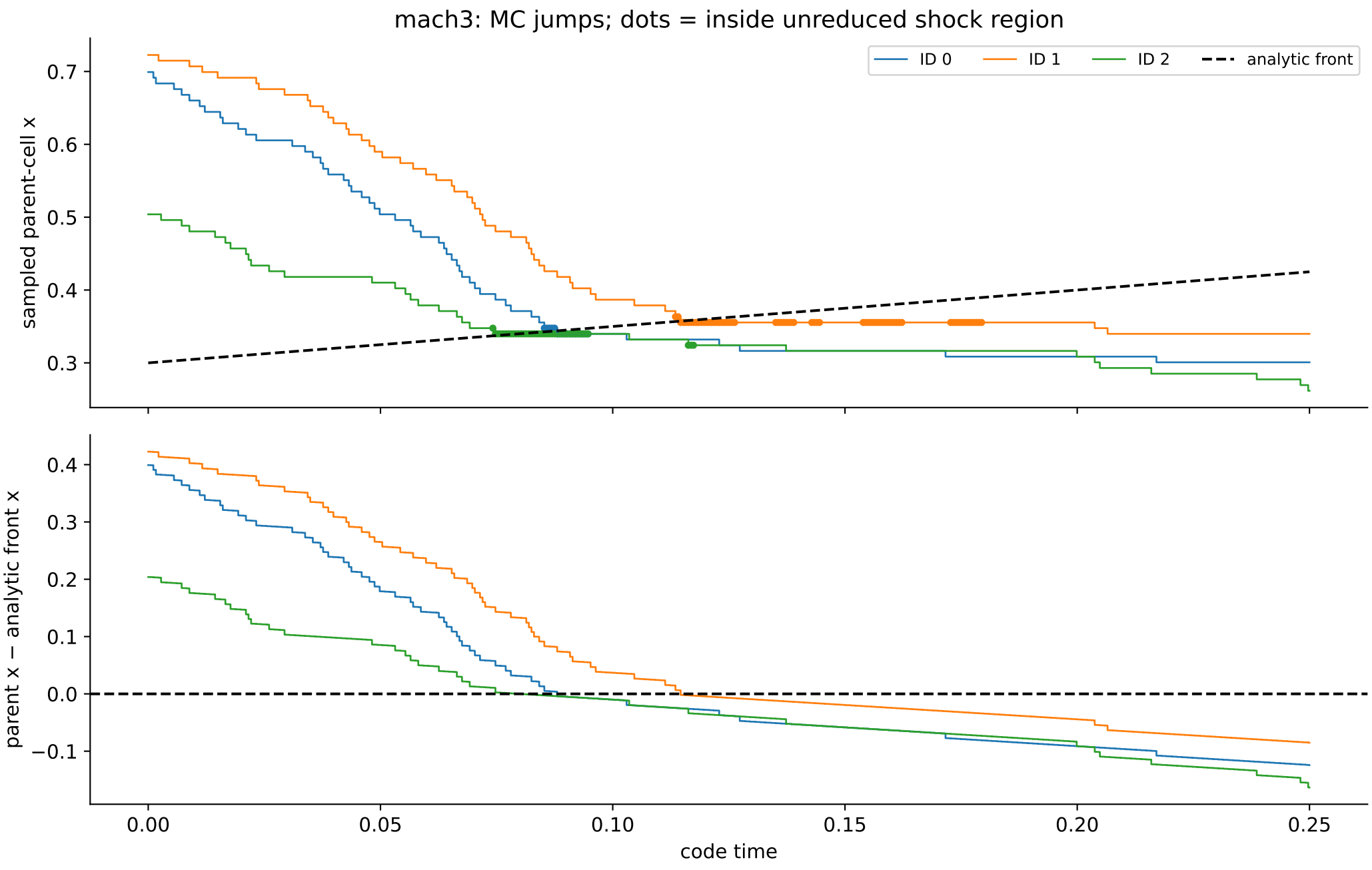
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0167 to +0.0103 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 11 | mach5 parents

Test setup

Input Mach 5, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

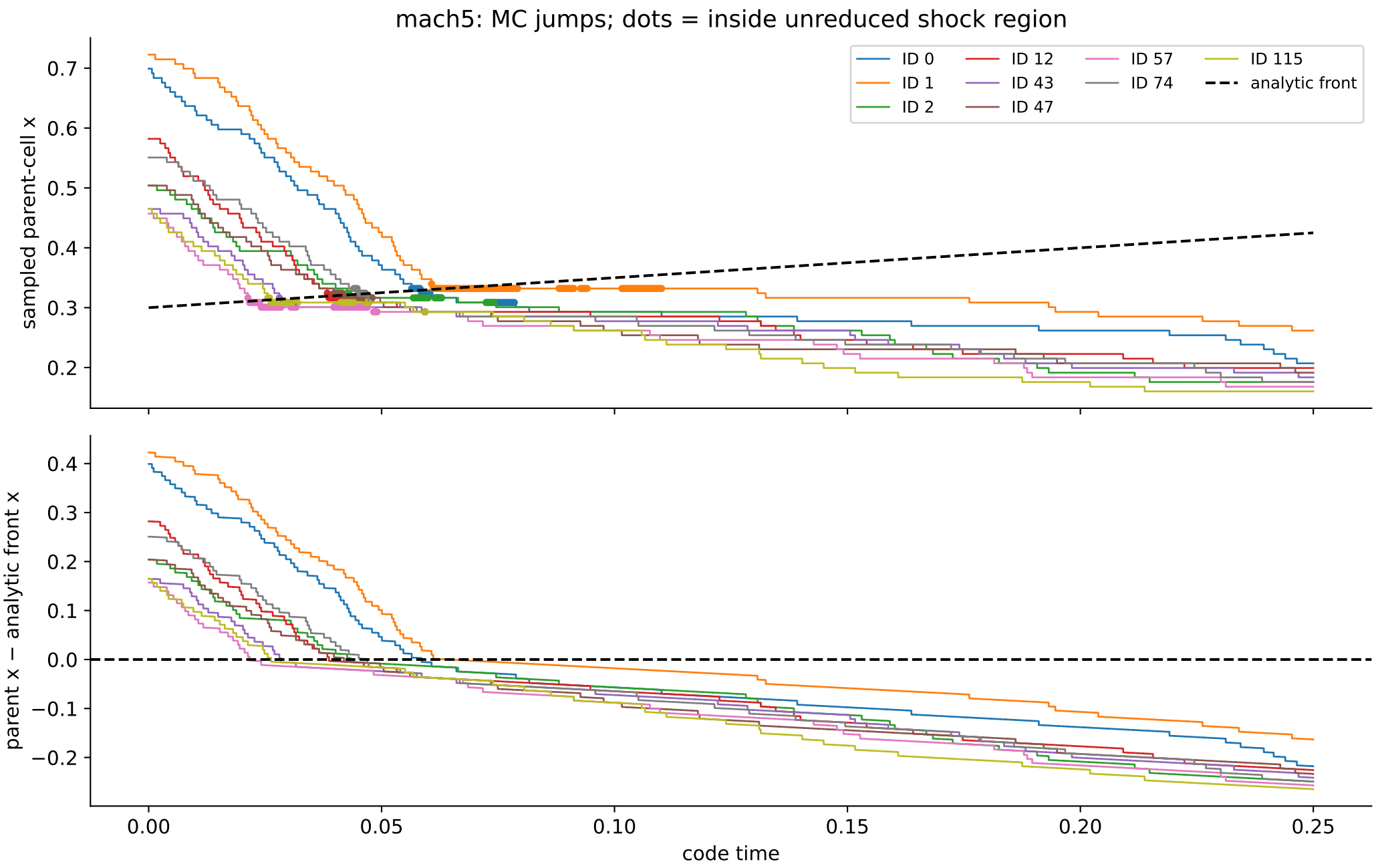
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

134 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -3.9657 to +0.0594 (estimate minus input Mach). 6 particles have a second episode; scientific interpretation remains pending.

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 12 | boost2 parents

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(2.5)t. The plotted IDs are selected samples.

How to read the plot

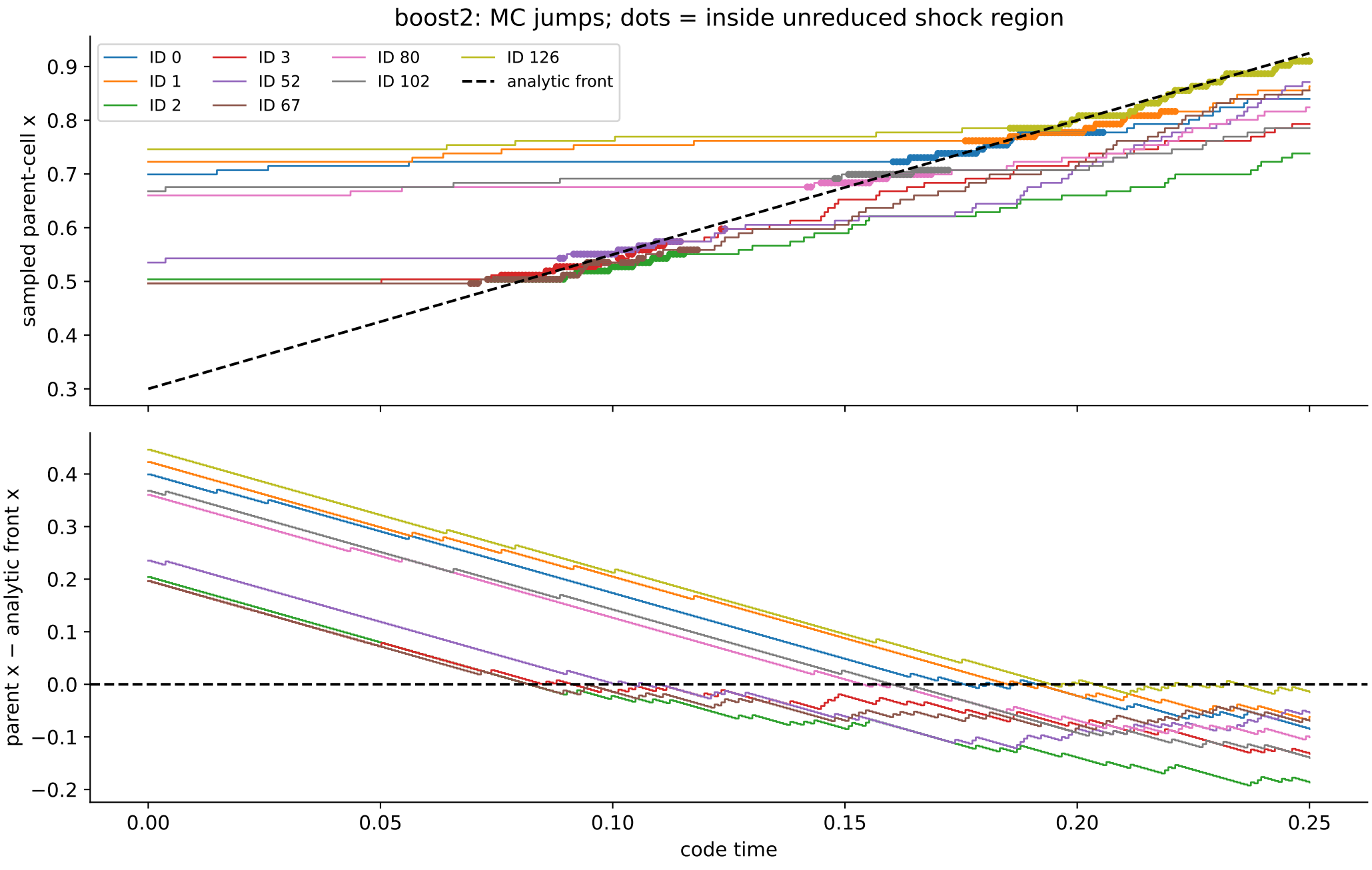
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

132 completed numerical episodes for 127 unique particles; 1 suspended. First-inside pressure-Mach error is -0.1091 to -0.0771 (estimate minus input Mach). 5 particles have a second episode; scientific interpretation remains pending.

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 13 | stationary parents

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0)t. The plotted IDs are selected samples.

How to read the plot

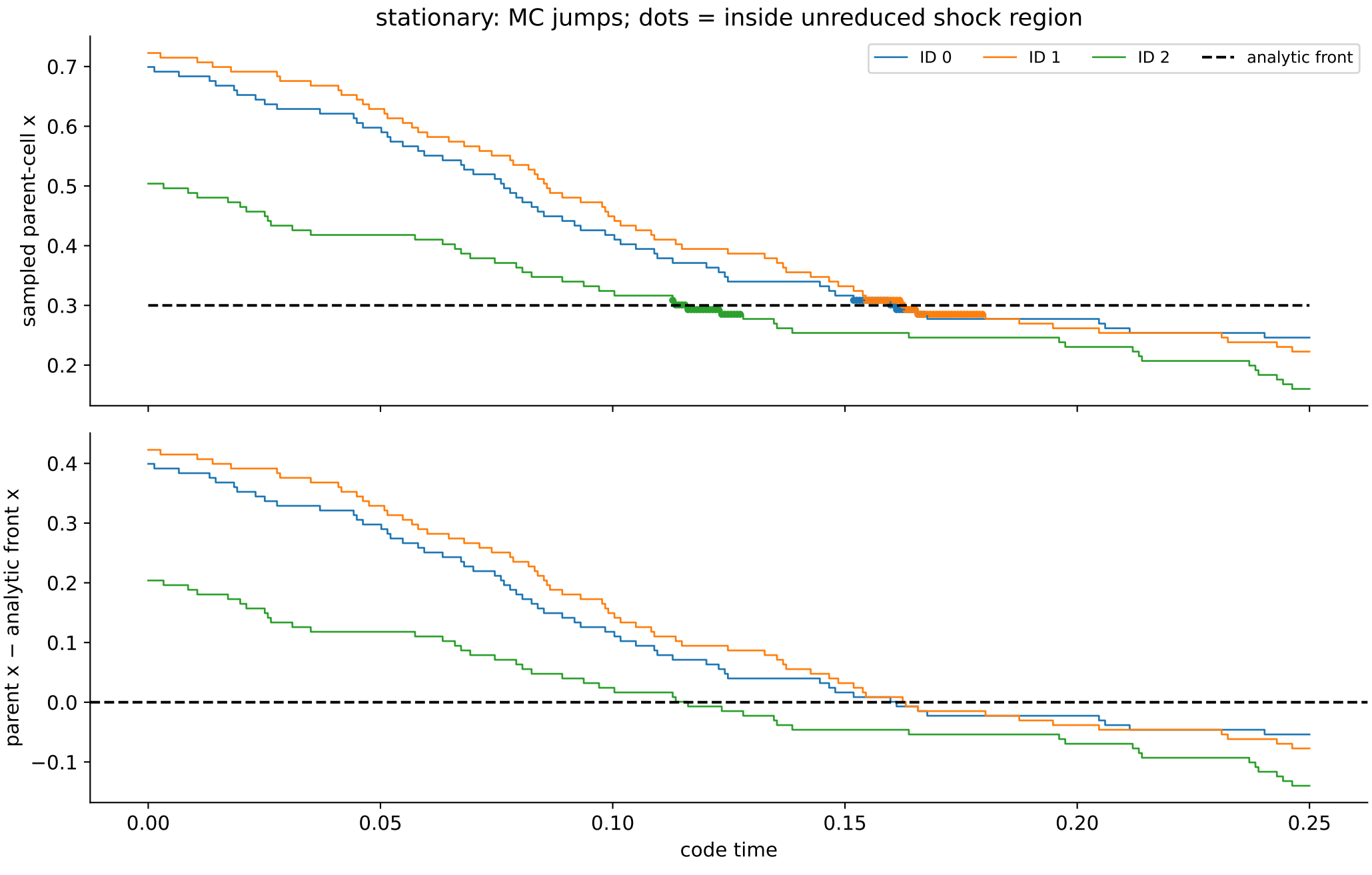
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0006 to +0.0004 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 14 | reflected parents

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.7+(-0.5)t. The plotted IDs are selected samples.

How to read the plot

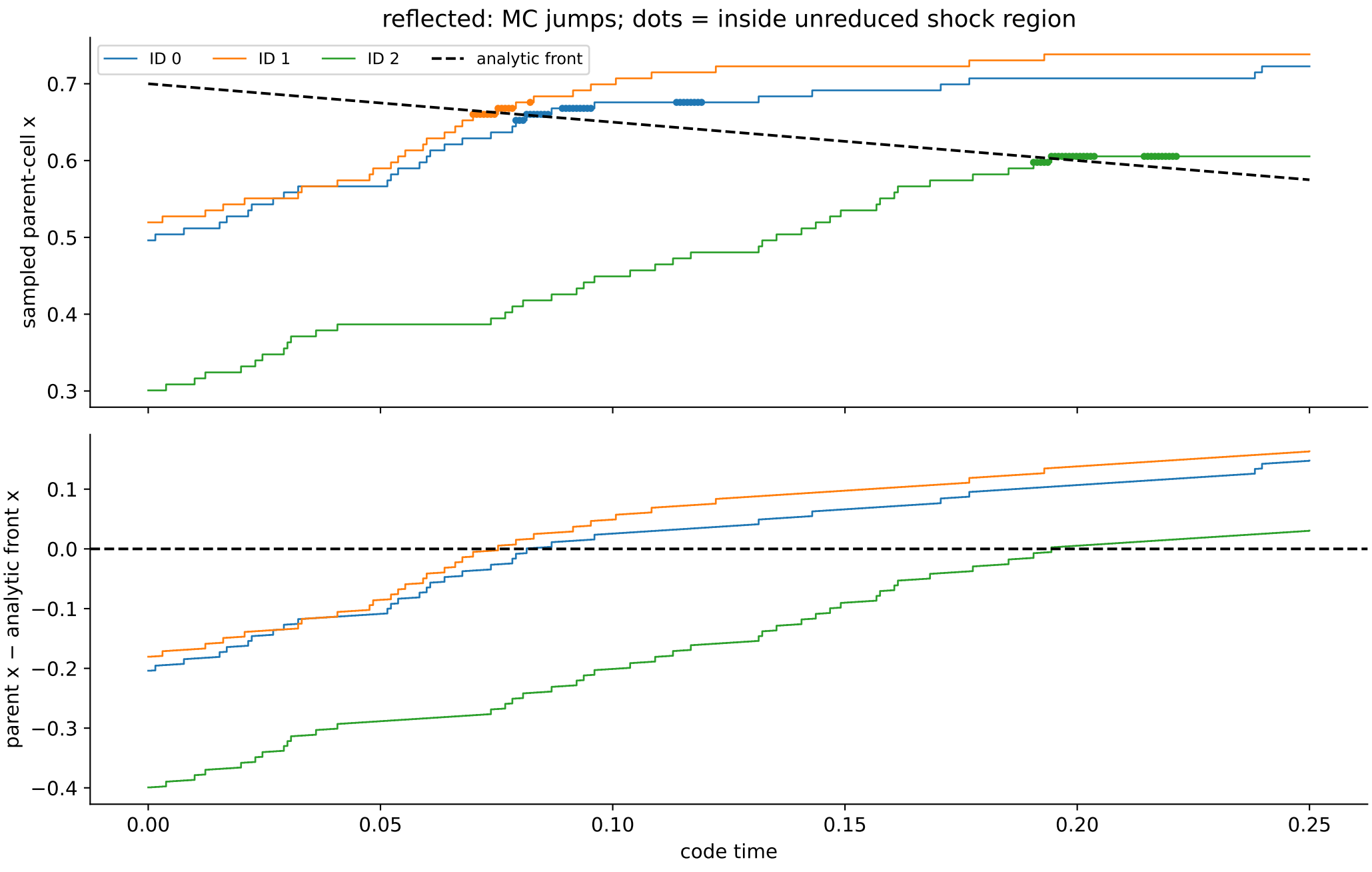
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0164 to +0.0120 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 15 | nx64 parents

Test setup

Input Mach 2, 64 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

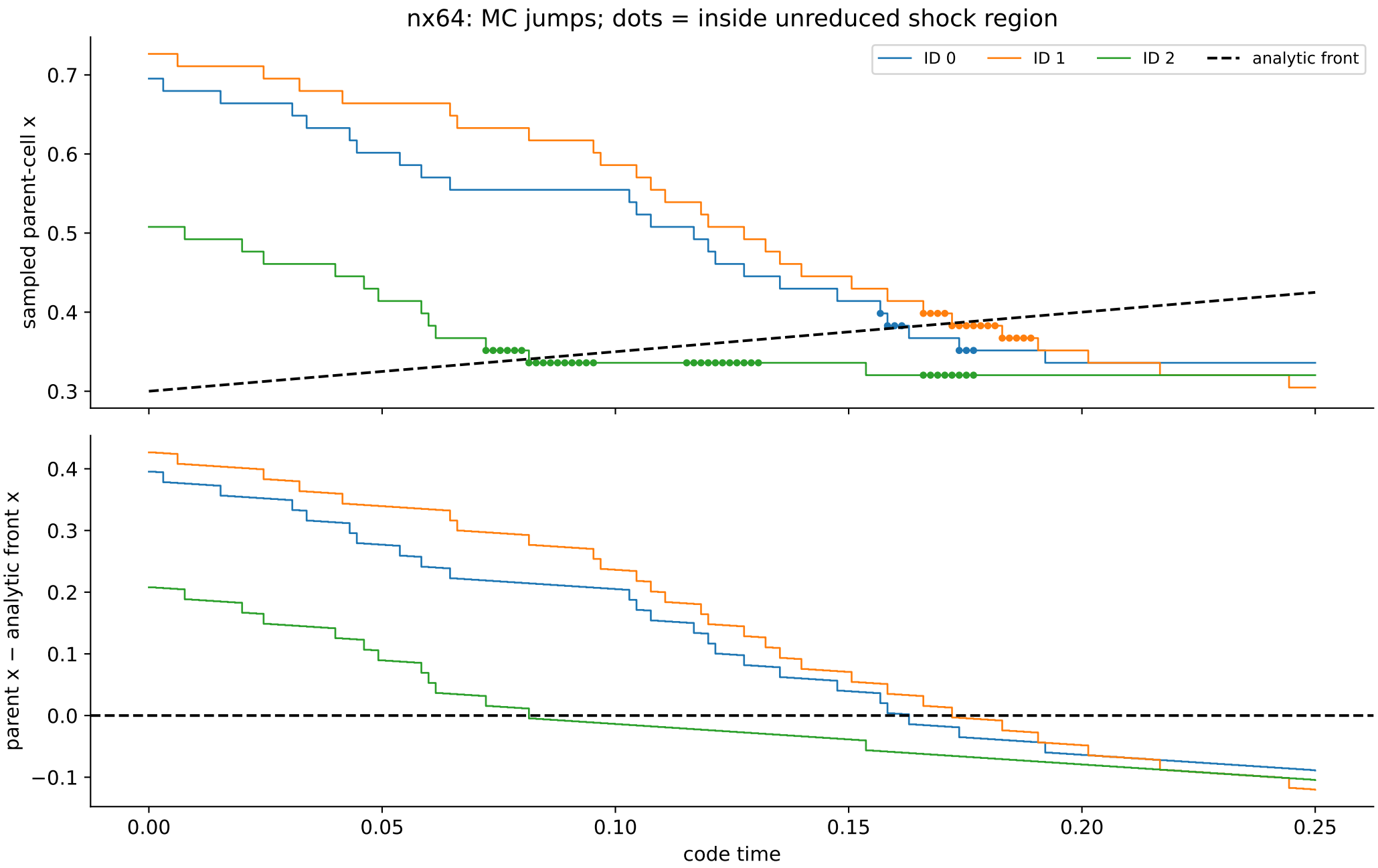
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0163 to +0.0120 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 16 | nx256 parents

Test setup

Input Mach 2, 256 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

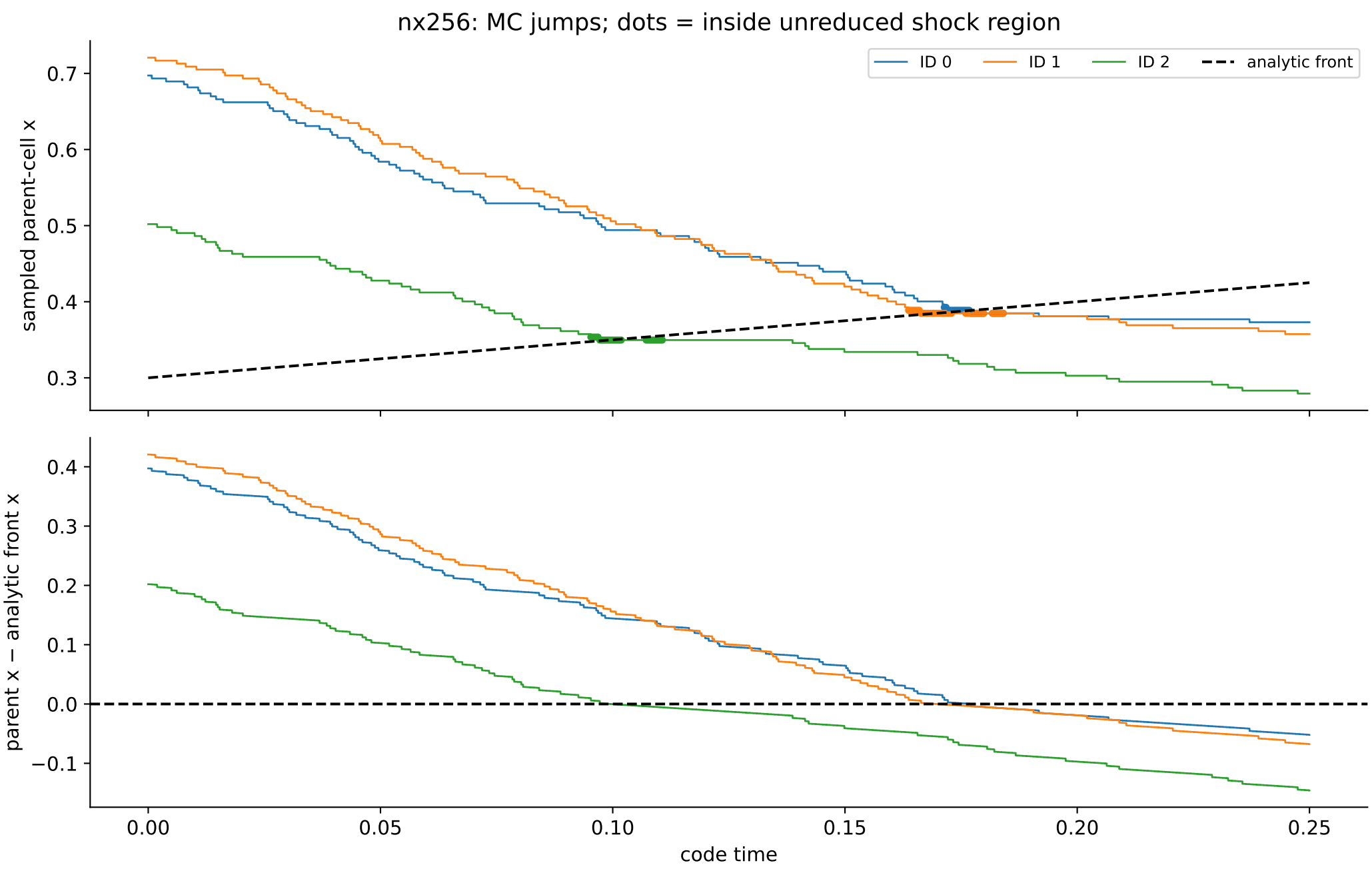
Top: sampled cell-center x versus time, with inside-region dots and the known dashed front. Bottom: parent x minus analytic front x. Stair steps show discrete cell hops; selected extra-episode/censored IDs are included.

Measured result

128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0164 to +0.0118 (estimate minus input Mach).

What this establishes

MC sample sequences are not continuous material paths. A numerical re-entry need not represent a distinct physical shock. No exact crossing time is assigned, and a suspended encounter is retained at cutoff.



Test 17 | seed11 strength time

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

Measured result

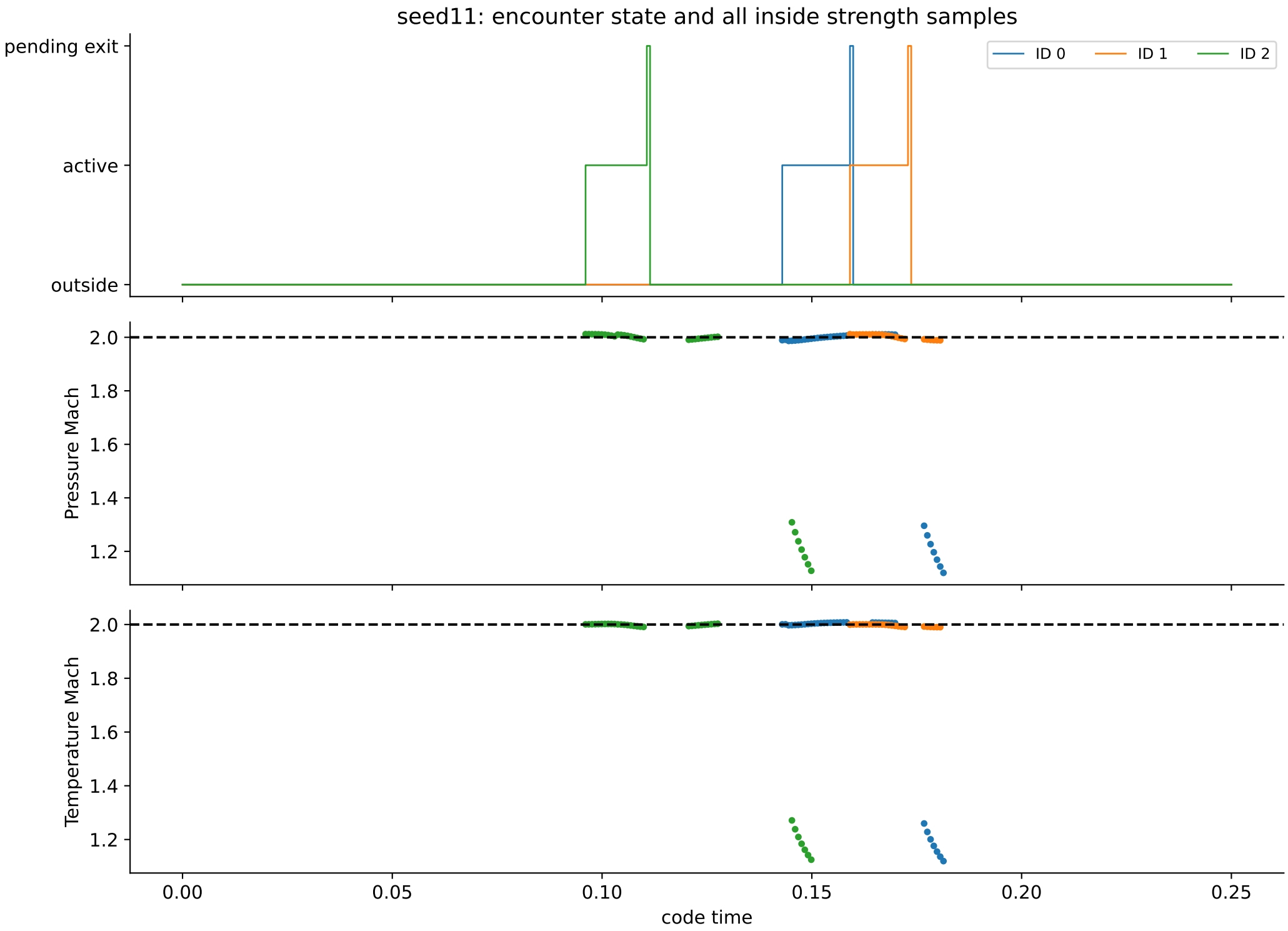
128 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -0.0163 to +0.0118 (estimate minus input Mach).

How to read the plot

Top: outside/active/pending-exit tracker state. Lower panels: pressure and temperature Mach for every valid inside observation. Dashed lines mark input Mach; gaps mean no valid inside strength sample.

What this establishes

The lower panels include follow-up samples, whereas each emitted event uses its first valid inside sample as representative. Fixed sampling endpoints are not certified undisturbed plateaus. Strength and physical passage acceptance remain pending.



Test 18 | mach5 strength time

Test setup

Input Mach 5, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(0.5)t. The plotted IDs are selected samples.

How to read the plot

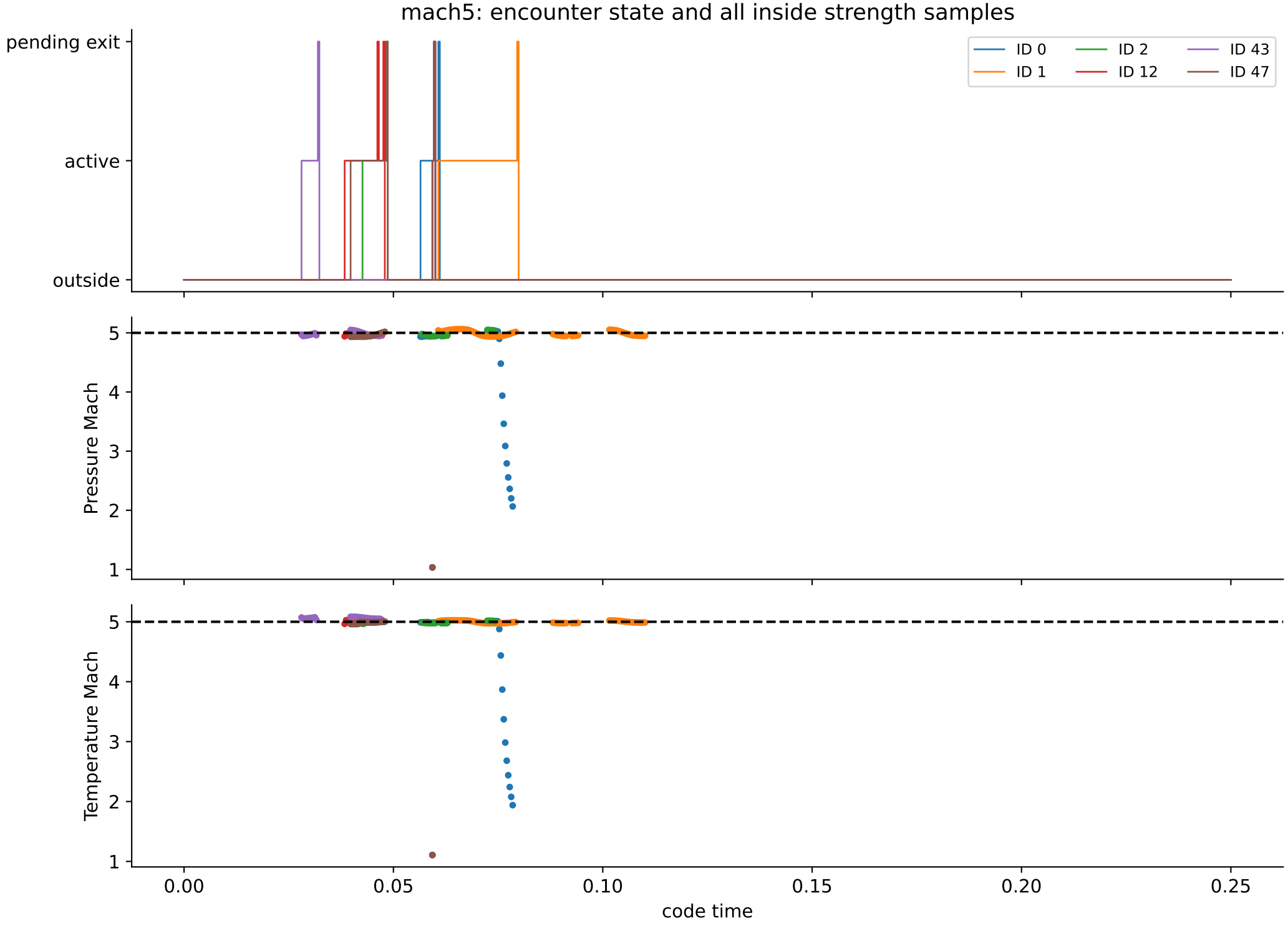
Top: outside/active/pending-exit tracker state. Lower panels: pressure and temperature Mach for every valid inside observation. Dashed lines mark input Mach; gaps mean no valid inside strength sample.

Measured result

134 completed numerical episodes for 128 unique particles; 0 suspended. First-inside pressure-Mach error is -3.9657 to +0.0594 (estimate minus input Mach). 6 particles have a second episode; scientific interpretation remains pending.

What this establishes

The lower panels include follow-up samples, whereas each emitted event uses its first valid inside sample as representative. Fixed sampling endpoints are not certified undisturbed plateaus. Strength and physical passage acceptance remain pending.



Test 19 | boost2 strength time

Test setup

Input Mach 2, 128 x 4 cells, 128 MC tracers, gamma=1.4, CFL=0.3, t_limit=0.25. Analytic front: x=0.3+(2.5)t. The plotted IDs are selected samples.

How to read the plot

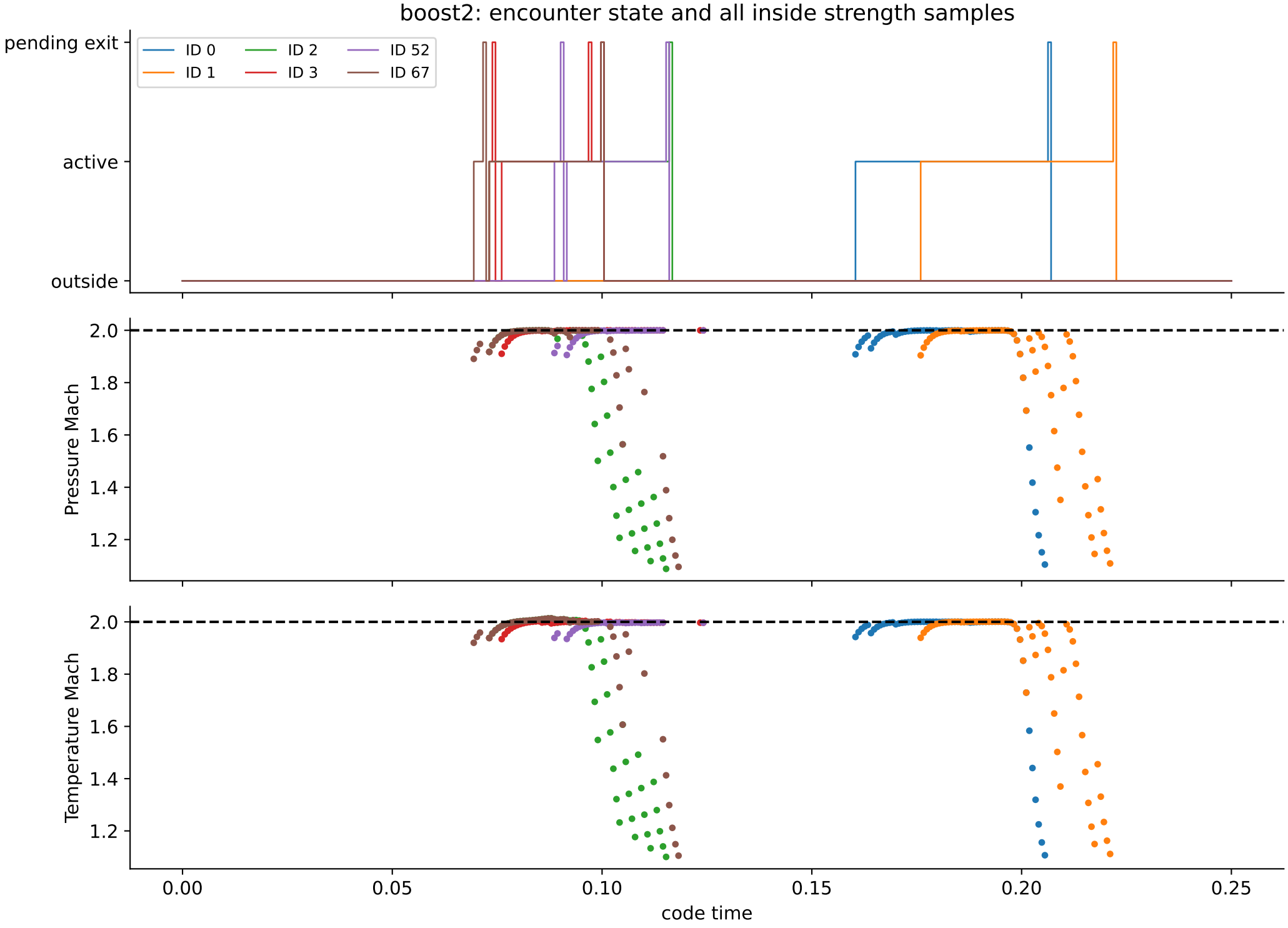
Top: outside/active/pending-exit tracker state. Lower panels: pressure and temperature Mach for every valid inside observation. Dashed lines mark input Mach; gaps mean no valid inside strength sample.

Measured result

132 completed numerical episodes for 127 unique particles; 1 suspended. First-inside pressure-Mach error is -0.1091 to -0.0771 (estimate minus input Mach). 5 particles have a second episode; scientific interpretation remains pending.

What this establishes

The lower panels include follow-up samples, whereas each emitted event uses its first valid inside sample as representative. Fixed sampling endpoints are not certified undisturbed plateaus. Strength and physical passage acceptance remain pending.



Test 20 | episode counts

Test setup

21 held-out runs: eight Mach 2 seeds, four other Mach numbers, gamma, front speed/direction, resolution, CFL and one thin-3D variation; 128 initial particles each.

How to read the plot

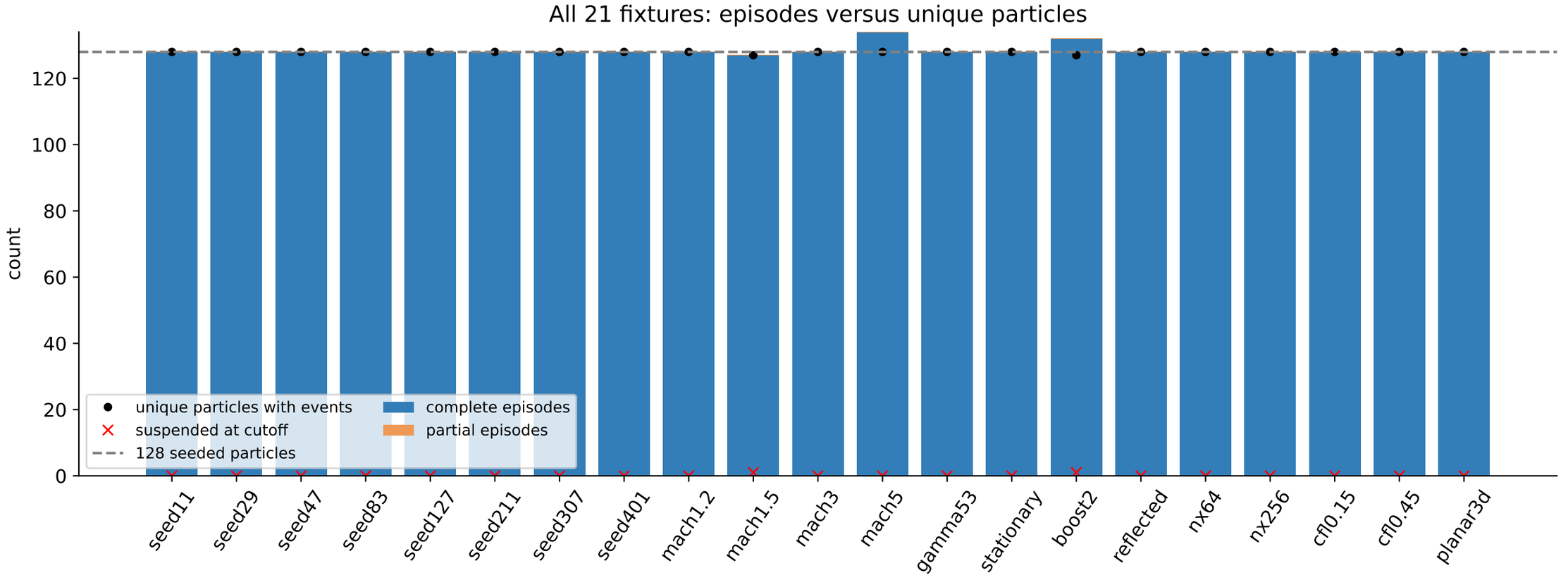
Blue bars count completed numerical episodes; orange adds partial episodes. Black dots count unique IDs with events; red crosses count suspended views. The dashed reference is 128 initially seeded particles.

Measured result

Every Mach 2 seed gives 128/128. Mach 5 gives 134 episodes/128 IDs. Boosted Mach 2 gives 132/127 plus one suspended. Mach 1.5 gives 127 plus one suspended. Other held-out cases give 128/128.

What this establishes

Successful execution is not scientific acceptance. Extra episodes violate the simple single-front counting expectation; unfinished encounters are censored evidence, not discarded failures.



Test 21 | mach errors

Test setup

First-inside representative strengths from all emitted events in the 21 held-out cases, compared with independently prescribed input Mach.

How to read the plot

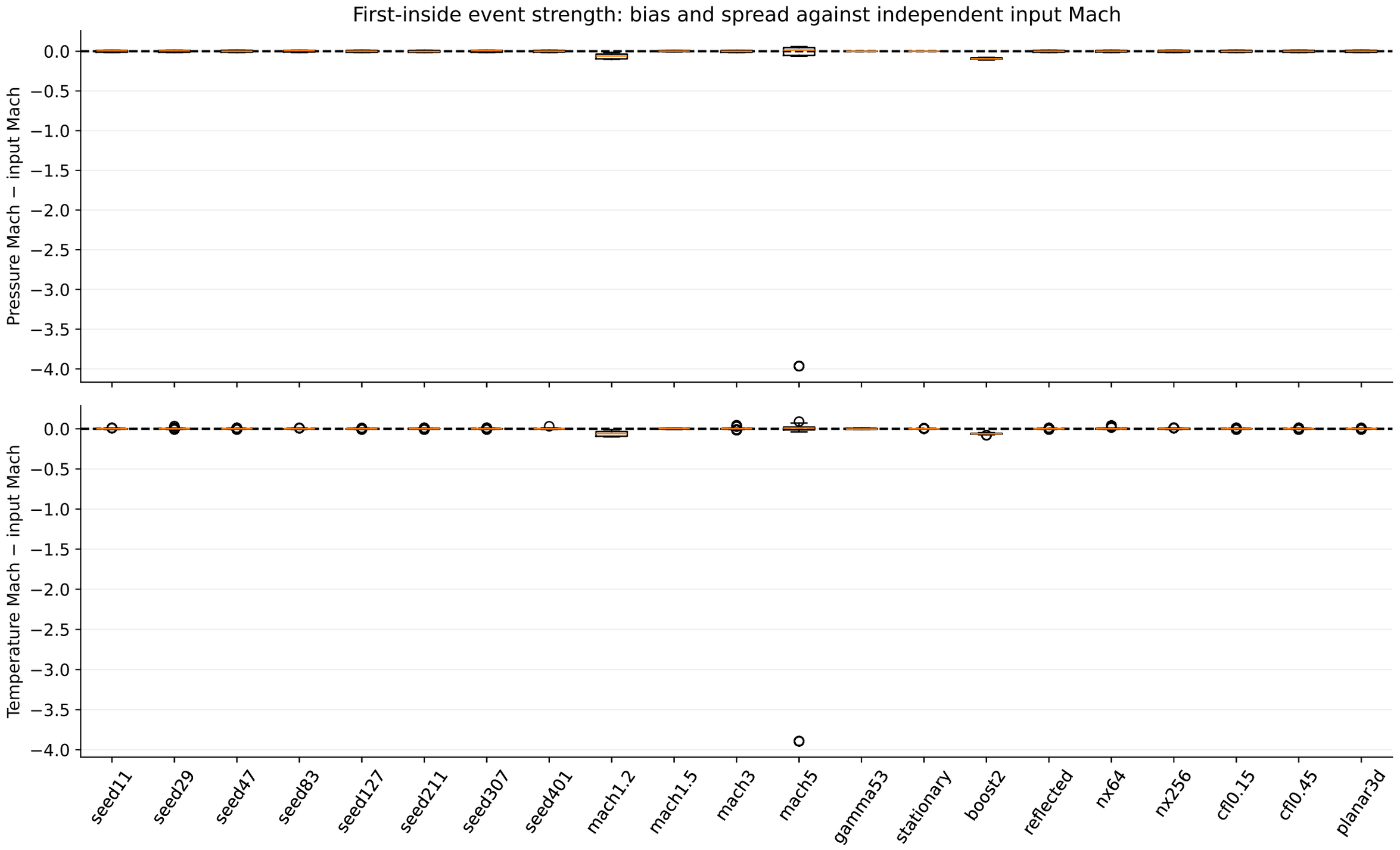
Top boxes use pressure Mach; bottom boxes use temperature Mach. The y axis is estimated minus input Mach. Zero is unbiased. Boxes, whiskers and outliers retain every emitted event, including weak secondary detections.

Measured result

Pressure-Mach errors: seed-11 Mach 2, -0.0163 to +0.0118; Mach 1.2, -0.1031 to -0.0157; Mach 5, -3.9657 to +0.0594; boosted Mach 2, -0.1091 to -0.0771.

What this establishes

A box plot diagnoses bias/spread, not an approved error budget. No outliers were removed, and the full vertical range intentionally exposes strong-case failures. Fixed sampling endpoints are not certified undisturbed plateaus. Strength and physical passage acceptance remain pending.



Test 22 | timing bracket widths

Test setup

Compare x resolutions 64/128/256 and CFL 0.15/0.3/0.45 for the same Mach 2 planar test, measuring each event's recorded entry and exit bracket width.

How to read the plot

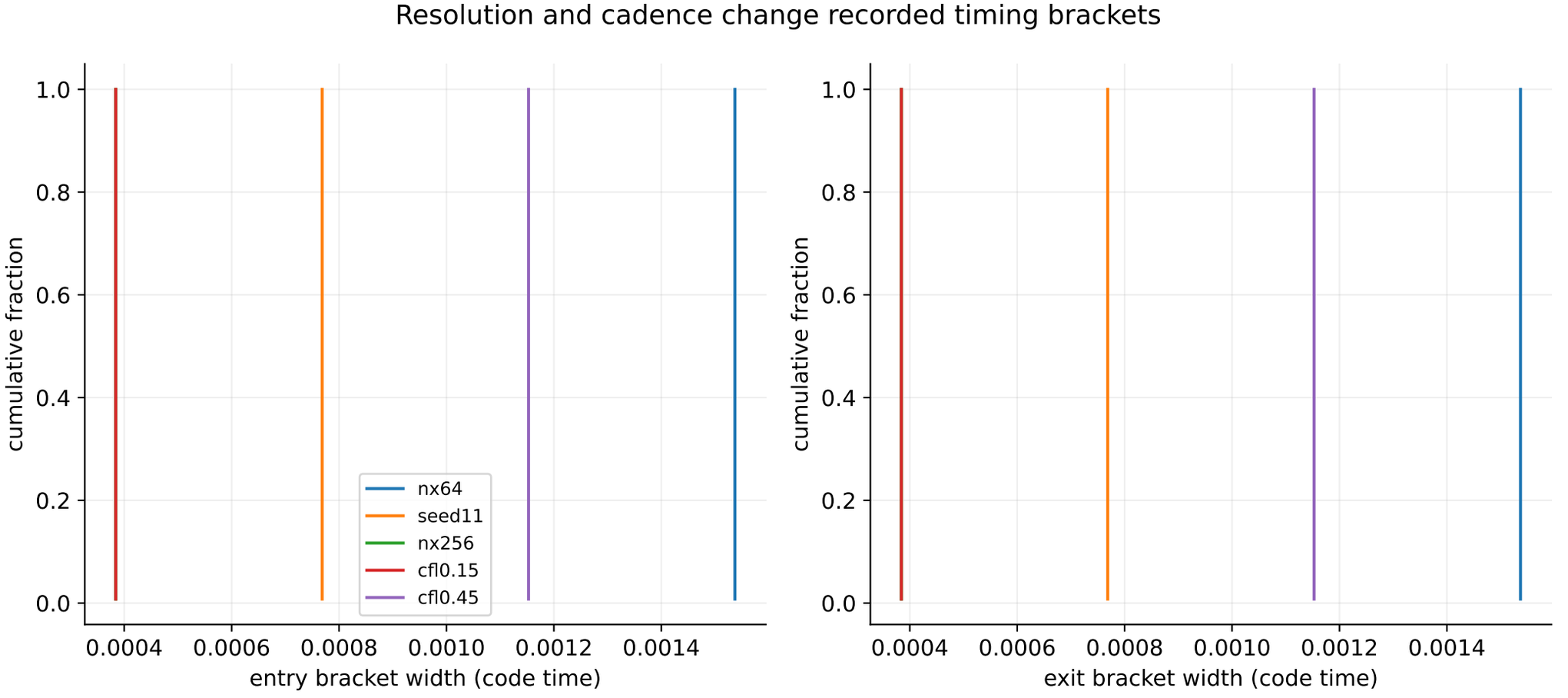
Left: entry widths; right: exit widths. Each cumulative curve shows the fraction of events below a width. A left shift means tighter observation brackets. Coincident curves can overlap.

Measured result

Largest recorded entry widths: nx64 0.00153706, baseline 0.000768531, nx256 0.000384266 code-time units. All five cases record 128 episodes for 128 IDs.

What this establishes

Widths describe adjacent mask observations, not errors against an exact material crossing. Halving CFL and doubling resolution can yield similar cadence; that does not prove converged physical passage histories.



Test 23 | encounter brackets

Test setup

Selected particle IDs in baseline Mach 2, Mach 5 and boosted Mach 2, including IDs with rearmed episodes and terminal suspended evidence.

How to read the plot

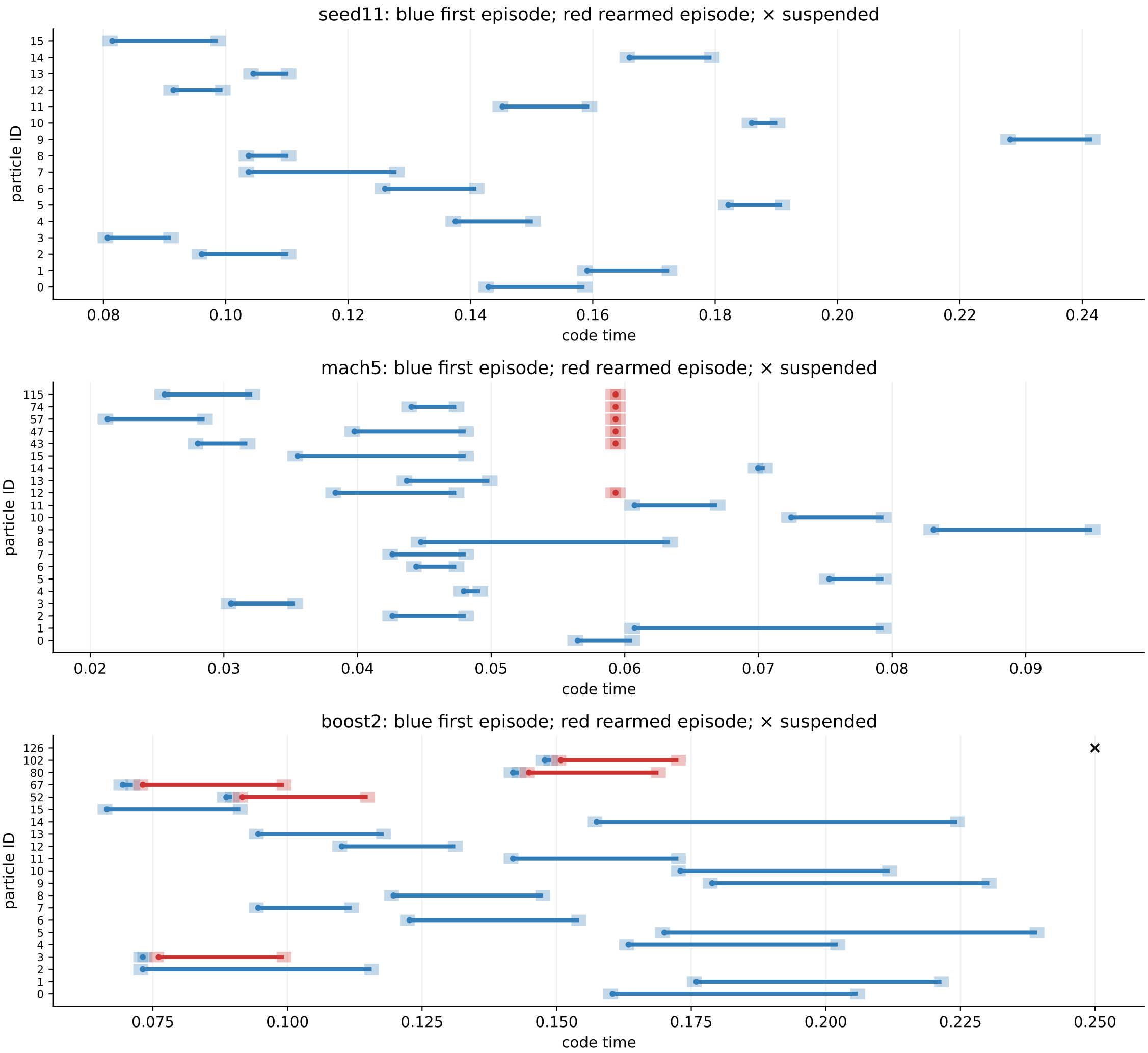
Blue is the first episode; red is a later episode for the same ID. Pale thick ends are entry/exit brackets. Solid segments join first and last inside observations; a black cross denotes a suspended terminal view.

Measured result

Baseline has one event per ID. Mach 5 has six IDs with a second episode; the boosted case has five and one suspended encounter. The tracker preserves these distinct identities and timing evidence rather than hiding them.

What this establishes

A solid segment is not exact physical residence: flicker can be merged, and MC returns are stochastic. Brackets limit sampled region transitions, not exact material shock crossings.



Test 24 | uniform transport

Test setup

Uniform periodic flow, velocity u=1, 64 x 4 cells, 1,024 tracers, t_limit=0.25. The independent reference uses per-step Bernoulli probabilities p=u*dt/dx.

Measured result

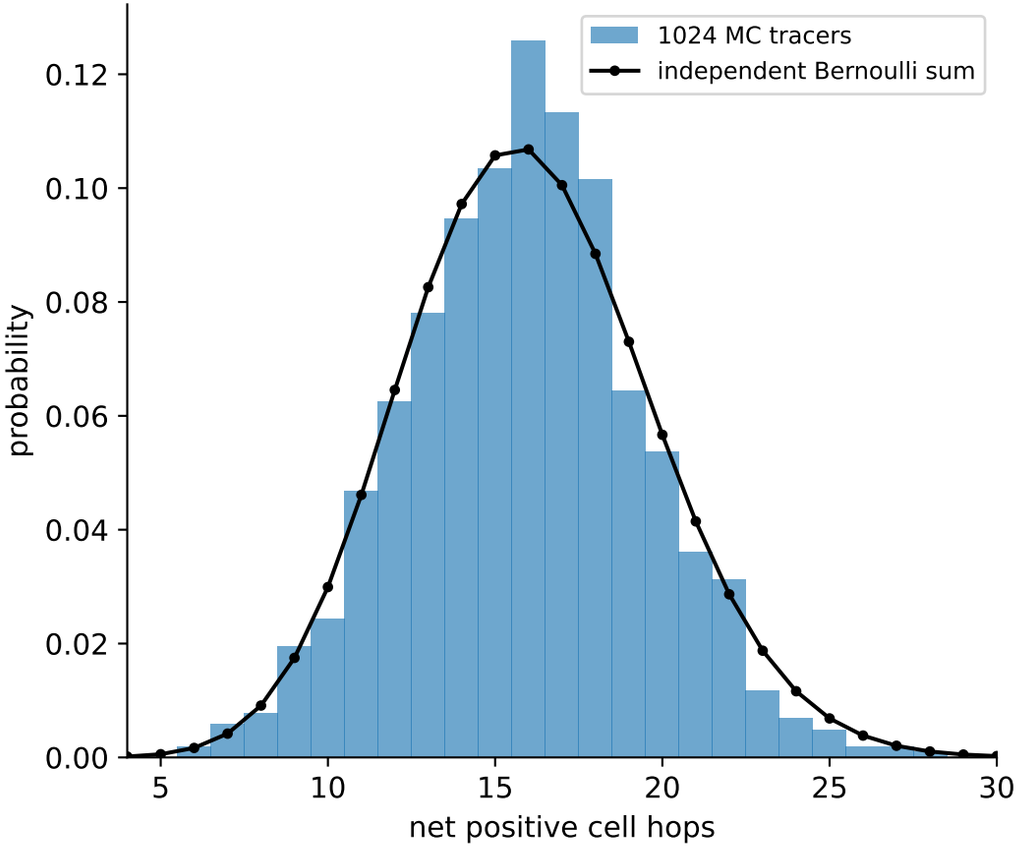
Expected mean 16.0000, measured 15.8965; expected variance 13.8061, measured 12.3314. Mean/variance deviations are -0.89/-2.40 standard errors; both pass.

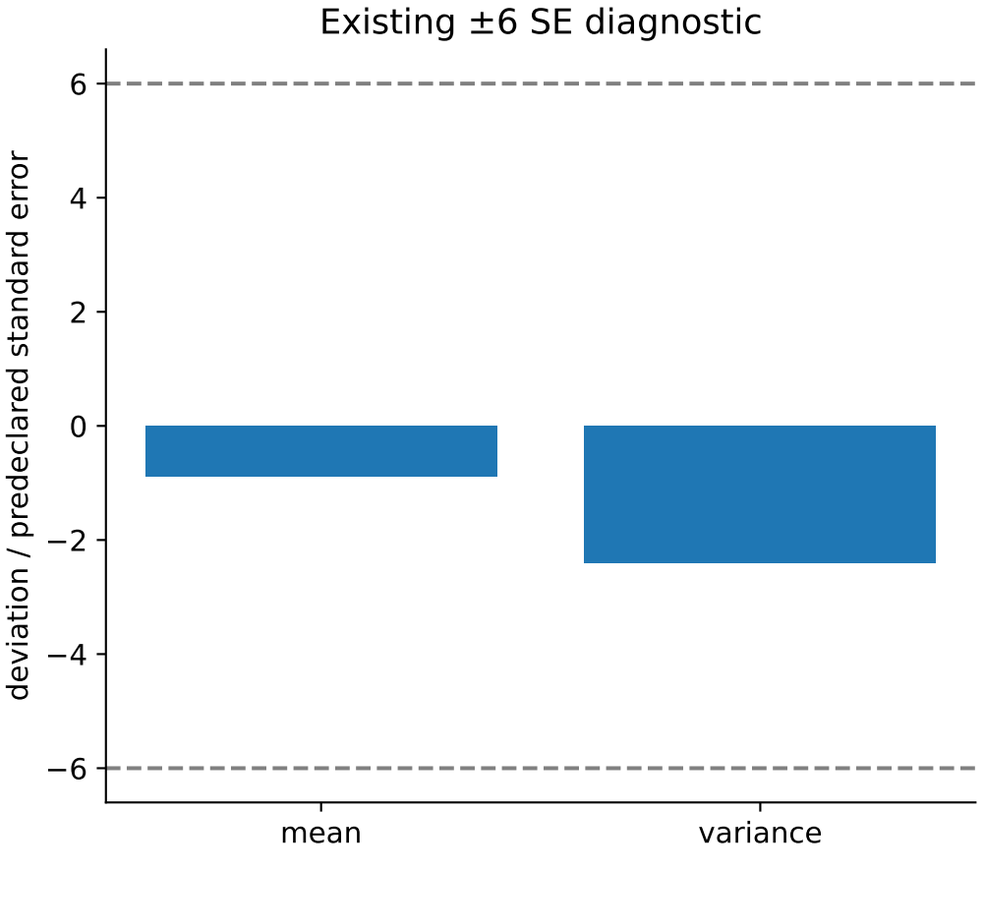
How to read the plot

Left: measured net-hop histogram versus the probability distribution of the Bernoulli sum. Right: mean and variance differences measured in their predeclared standard errors, with the existing +/-6 diagnostic lines.

What this establishes

This is a fixed-seed ensemble transport check, not proof of RNG independence at research scale. The inherited additive RNG key has alias/correlation questions; no deterministic per-particle path oracle is used.





Test 25 | negative controls

Test setup

Uniform flow, a contact with density ratio 2:1 at constant pressure and velocity, and a zero-particle run. A contact should not be mistaken for a shock.

How to read the plot

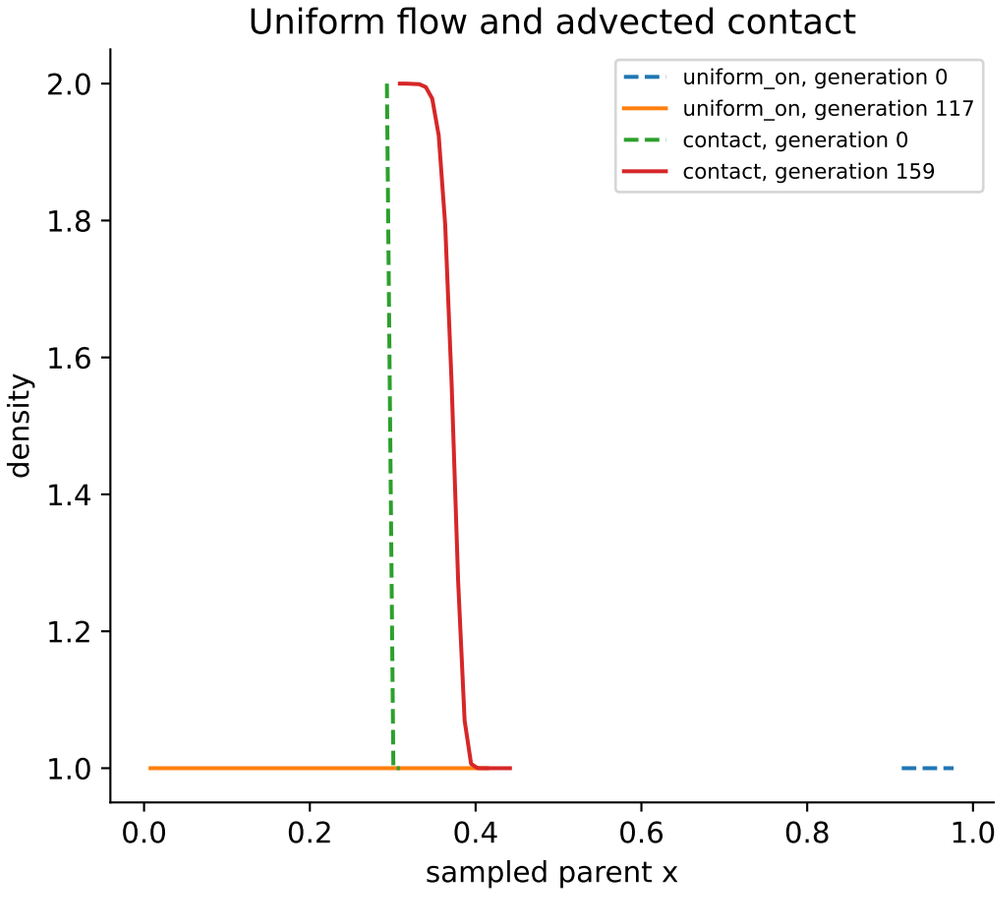
Left: sampled density versus parent-cell x at the first and last generations for uniform/contact runs. Right: emitted event counts for all three controls.

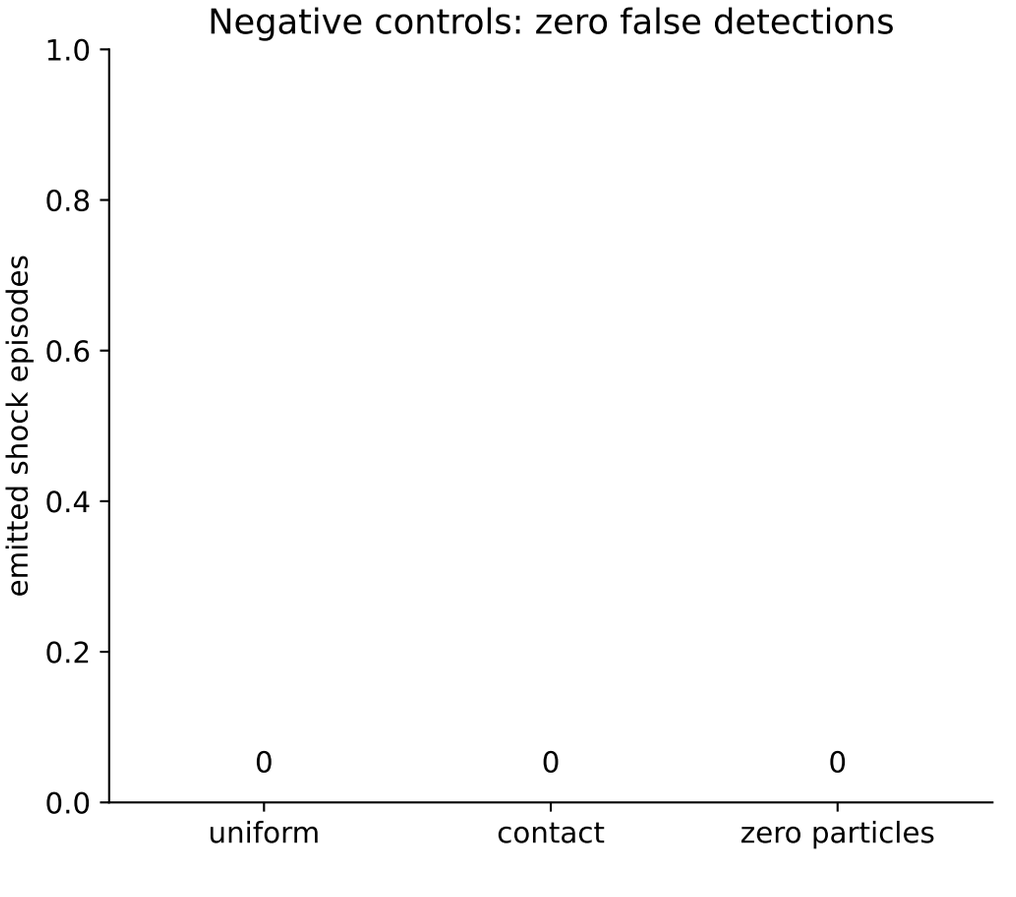
Measured result

All three emit zero shock episodes. Tracking-on/off uniform and planar runs also preserve exact native particle observations and byte-identical 17-digit conserved slices in the separate non-interference checks.

What this establishes

These controls check specific false detections and empty-population behavior. They do not bound the false-positive rate in turbulence or complex shock geometry.





Test 26 | restart agreement

Test setup

Checkpoint the baseline run in outside, active and pending-exit phases (generations 1, 61 and 68), then resume with identical source/configuration/partition.

How to read the plot

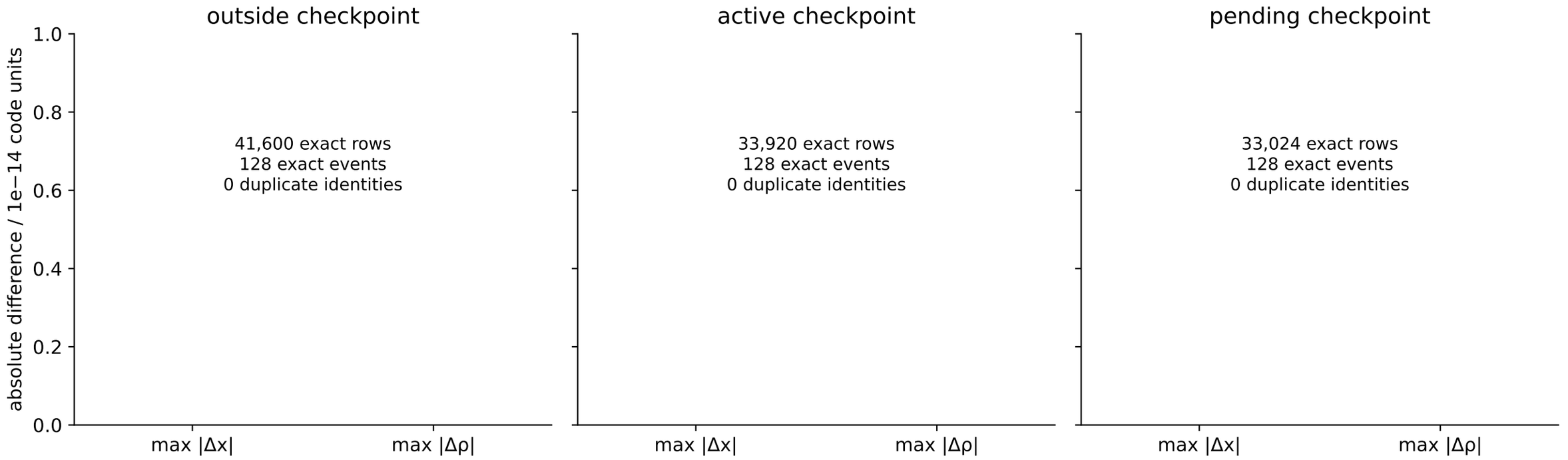
Each panel plots maximum absolute position and density differences from uninterrupted particle observations. The display unit is 1e-14 code units; comparisons use exact equality, not a 1e-14 tolerance.

Measured result

All differences are exactly zero. Outside/active/pending resumes match 41,600 / 33,920 / 33,024 rows, respectively, and each preserves all 128 canonical events with no duplicate identities.

What this establishes

Requires the matching implementation, native ABI, rank count and mesh partition plus retained event prefix. No cross-rank, all-rank atomic transaction or power-loss/storage-fault qualification is claimed.



Test 27 | runtime cost

Test setup

Uniform flow, 1,024 particles, t_limit=0.1, grids 64 x 4 and 256 x 4. Three trials per off/tracking/trace mode; order is rotated.

How to read the plot

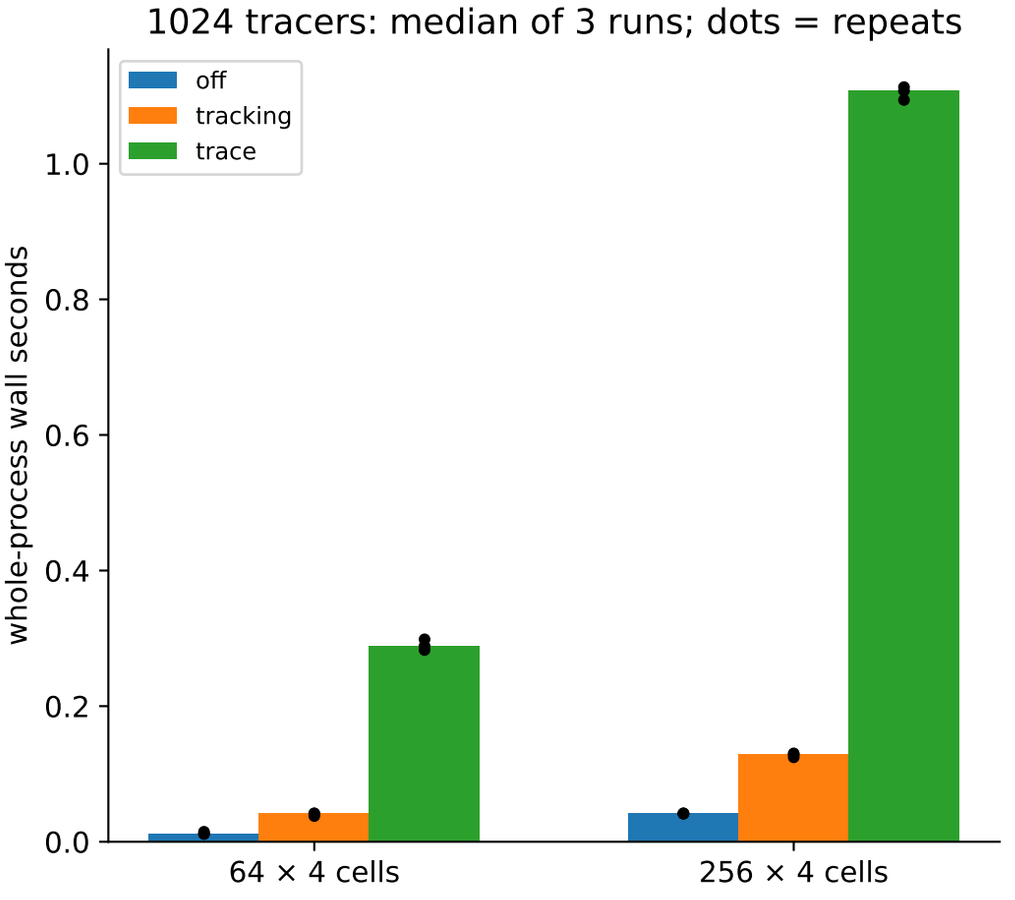
Left: median whole-process wall seconds, with dots for actual repeats. Right: runtime divided by the matching tracking-off median. Blue is off, orange tracking with trace off, and green full diagnostic traces.

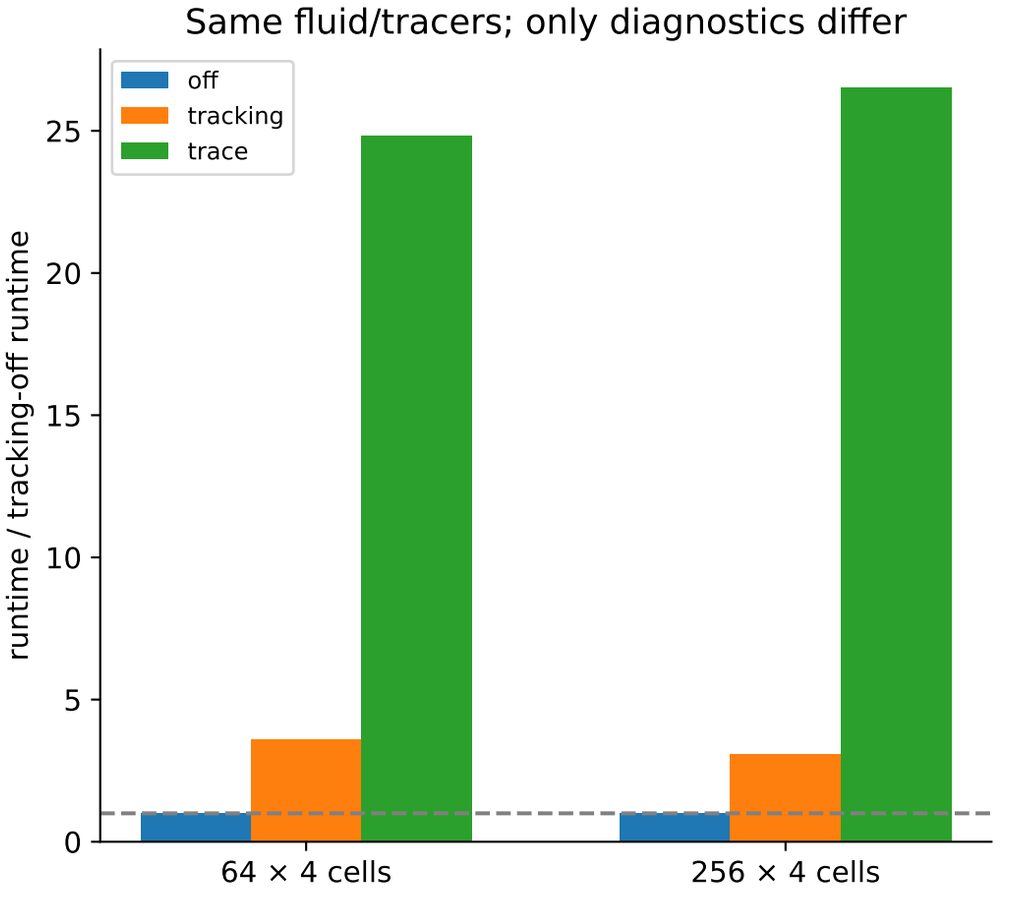
Measured result

64 x 4: off 0.01163 s, tracking 0.04171 s (3.59x), trace 0.28880 s (24.84x). 256 x 4: off 0.04174 s, tracking 0.12862 s (3.08x), trace 1.10772 s (26.54x).

What this establishes

Startup, ordinary output and durable empty event chunks are included. Full traces are expensive small-job qualification diagnostics. These timings do not establish sustained RUSTY/Frontier or MPI/GPU performance.





Test 28 | output cost

Test setup

The same two-mesh, 1,024-particle benchmark, inspecting shock diagnostic directories after each off/tracking/trace run.

How to read the plot

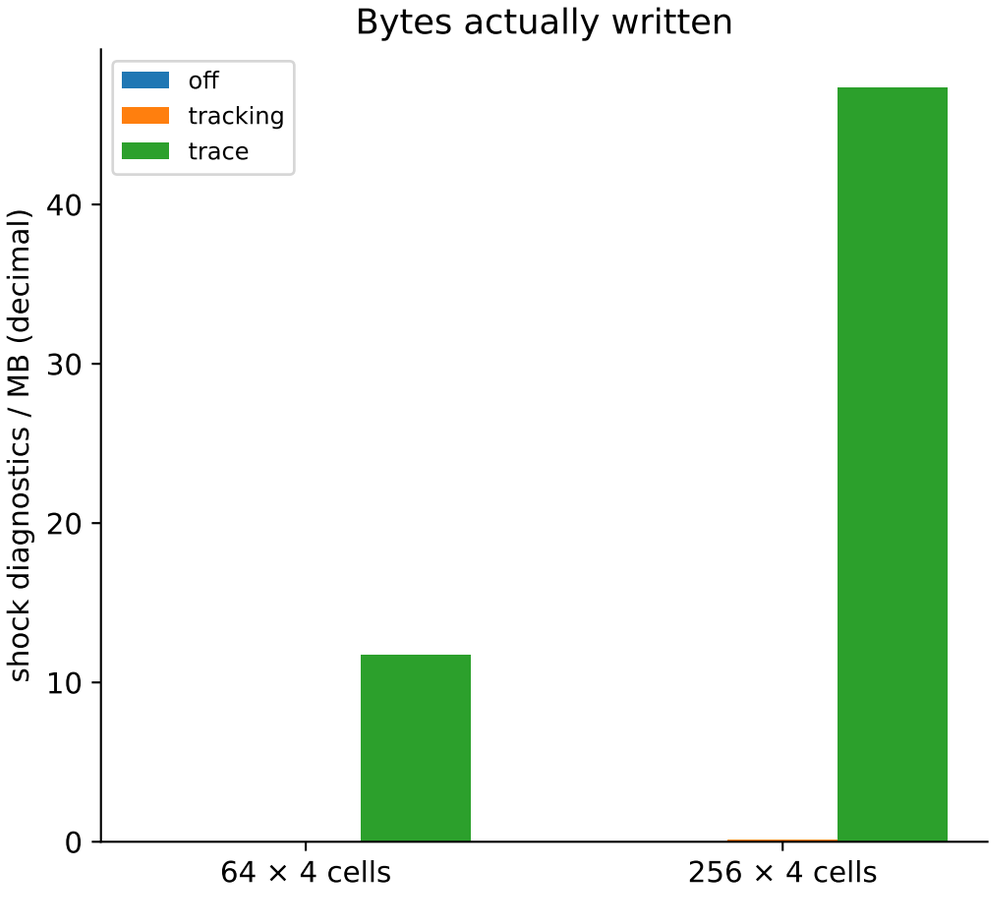
Left: bytes written in decimal MB; right: file count. Bars use the same three diagnostic modes as the runtime plot. Counts include immutable per-generation chunks and optional observation/transfer CSVs.

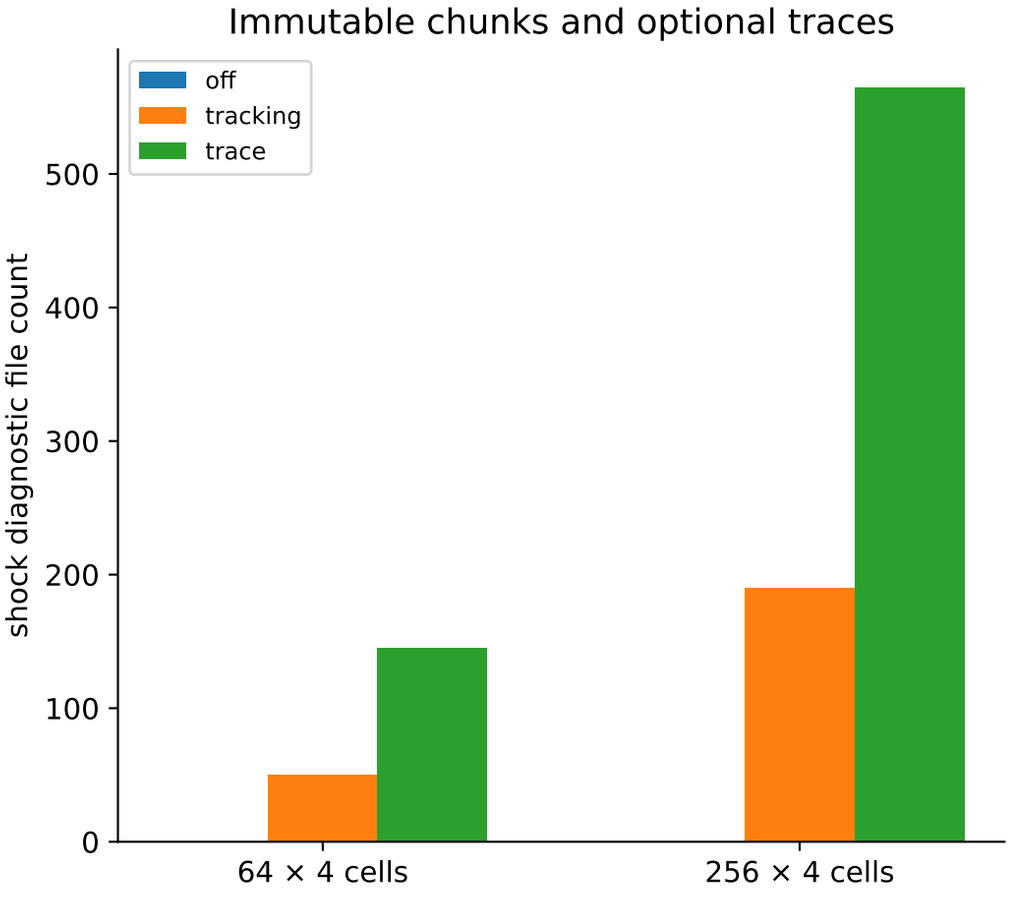
Measured result

Trace mode writes 11.73 MB / 145 files on 64 x 4, and 47.37 MB / 565 files on 256 x 4. Tracking-off writes no shock diagnostic directory.

What this establishes

Fluid and legacy thermo outputs are excluded from these totals. Empty event generations still require durable files. Resident memory and production-filesystem behavior were not measured.





Appendix A. The primary physical experiment

A translating planar shock is evolved in a thin two-dimensional domain. Independent Rankine-Hugoniot jump relations define its initial states and known speed. This is genuine 2D fluid output, but the planar setup does not qualify curved or intersecting shocks.

Setting	Pilot configuration
Domain and grid	x: 0 to 1; y: 0 to 0.1; 128 x 4 cells; no AMR.
Physics and numerics	Newtonian ideal-gas hydro, gamma=1.4; RK2; PLM; HLLC; CFL=0.3.
Shock	Mach 2; initial front x=0.3; speed=0.5; x_front(t)=0.3+0.5t.
Particles and duration	128 tracers seeded in x=0.45 to 0.75; t_limit=0.25.
Boundaries	Outflow along x; periodic along y.
Diagnostic settings	width=4 cells; min_mach=1.1; capacity=4096; trace=true. These are pilot settings, not approved scientific tolerances.

Independent conservation reference

For upstream Mach M and gamma g, the density ratio is (g+1) M^2 / ((g-1) M^2+2), and the pressure ratio is (2g M^2-(g-1))/(g+1). The fixture checks mass, momentum and energy conservation in the shock frame before running the simulation.

At M=2, g=1.4: density ratio=2.6666667 and pressure ratio=4.5. Upstream density and pressure are both 1. The downstream values are 2.6666667 and 4.5. Velocities are -1.8664319 upstream and -0.3874120 downstream.

The pressure and density plots therefore have a reference independent of the detector. P/rho is a temperature proxy in code units. The Eulerian density-midpoint location is a front diagnostic; it is not used to invent particle crossing times.

Why particles encounter this front

The front moves right while the upstream fluid and sampled tracers move predominantly left. Their histories meet the resolved shock region. Staying inside that region for many timesteps should create one numerical episode, not one event per timestep.

Appendix B. Engineering tests: what each checks

The recent integration suite has 20 subprocess runs. Some must succeed; others must abort with a specific diagnostic. All produced the expected outcome. Test IDs refer to the implementation plan and recorded checks.

Test / IDs	Method and recorded result
Non-interference CFG01, PHY08, SYS02	Run identical uniform and planar problems with tracking on/off. Particle observations are exact matches; 17-digit conserved slices are byte-identical.
Sampling INT06, INT08	Compare final-step shock samples with native particle thermo history. Generation, time, position and density agree exactly.
Negative controls CFG06, PHY09	Uniform flow, contact at constant pressure/velocity, and no particles. All emit zero events.
Local migration INT01, IO01	Use multiple local mesh blocks. All initial IDs survive; 113 particles change parent block. This is not an inter-rank runtime test.
Birth/censoring EVT02, EVT05	Seed initially or later inside the shock region. Missing valid pre-entry evidence is marked left-censored.
All-phase restart RST01, RST07	Checkpoint outside, active and pending-exit phases. Resume and compare exact particle rows and canonical events.
Damage/replay RST03/04, IO08	Remove/corrupt sidecars and damage retained history. Expected explicit rejection; immutable replay checks prevent conflicting event timelines.
Capacity / configuration IO03/04, CFG	Force capacity=1 and an overflow; request unsupported RK3. Both explicitly reject instead of silently dropping state or accepting unqualified integration.

The native encounter unit also passes state-transition checks. Separate earlier evidence contains 12 explicit-rejection runs, 4 replay runs, 18 actual-method boundary cases (with some simplified harness context), and 13 actual production domain-exit records. These categories must not be summed as independent physical shock passages.

Appendix C. Detector parity and independent transport

ShockIt parity: fidelity of the numerical port

53 fixtures compare the native detector against the frozen ShockIt local_pass reference. Shock-region masks must match exactly. Floating quantities use unchanged absolute and relative tolerances of 1e-12. All 53 pass.

The fixtures exercise constant fields, ideal shock jumps, gradients/normals, estimator thresholds and invalid inputs. Matching the reference validates the port; it does not independently certify that ShockIt or the encounter association is scientifically correct.

Uniform transport: an independent physical probability model

For resolved positive uniform velocity u, the expected cell-hop probability per step is p=u*dt/dx. Summing the Bernoulli hops gives independent reference mean sum(p) and variance sum(p*(1-p)). This uses physical flux probabilities, not the detector or RNG implementation.

Quantity	Reference	Measured
Particles / steps		1024 tracers / 117 steps
Mean cell hops	16.000000	15.896484
Hop variance	13.806057	12.331405
Mean / variance deviation	Existing +/-6 standard-error diagnostic	-0.89 / -2.40 standard errors

Both diagnostics pass. TRC11 also checks 119,808 actual transfer records: source parent, step start/end times, probabilities, random draw, selected direction and realized hop. Maximum probability discrepancy is 6.11e-16 against the unchanged 1e-12 tolerance.

This fixed-seed ensemble check is not proof of RNG independence at research scale. The inherited additive RNG key has alias/correlation questions that remain open. There is no deterministic per-particle material-path oracle.

Appendix D. Physical recovery and strength limitations

All 21 held-out fixtures executed. They vary eight Mach 2 seeds, Mach=1.2/1.5/3/5, gamma=5/3, stationary/boosted/reflected fronts, nx=64/256, CFL=0.15/0.45 and one thin-3D case. Run success is distinct from physical acceptance.

Fixture	Complete	Unique IDs	Suspended
Mach 2: each of eight seeds	128	128	0
Mach 1.2	128	128	0
Mach 1.5	127	127	1
Mach 3	128	128	0
Mach 5	134	128	0
Boosted Mach 2	132	127	1
Other gamma/speed/direction/resolution/CFL/thin-3D cases	128 each	128 each	0

Mach 5 gives six additional episodes for IDs 12, 43, 47, 57, 74 and 115. Boosted Mach 2 gives five additional episodes for IDs 3, 52, 67, 80 and 102, plus one suspended encounter. These are preserved limitations against the simple single-front expectation; stochastic returns and weak region lobes need scientific interpretation.

The boosted case adds 2 to the fluid and front velocities, retaining Mach 2. Its differing counts and bias expose frame dependence that needs qualification.

Case	Pressure-Mach error range
Mach 2, seed 11	-0.0163 to +0.0118
Mach 1.2	-0.1031 to -0.0157
Mach 5, all episodes	-3.9657 to +0.0594
Boosted Mach 2	-0.1091 to -0.0771

Error means estimated Mach minus prescribed Mach. Strength uses the first valid inside sample, with fixed endpoints that are not certified plateaus. Weak secondary episodes remain in the Mach 5 range. No outliers or tolerances were adjusted for acceptance.

Appendix E. Encounter timing, output and restart

An episode begins from valid unreduced-region membership, with an upstream-state entry guard where prior evidence exists. A first outside observation starts pending exit; two consecutive valid outside samples close/rearm. Earlier re-entry merges and flags flicker. Invalid samples cannot close an episode.

For valid pre/post evidence, entry bracket=[pre.time, first.time] and exit bracket=[last.time, post.time]. These bound adjacent sampled membership changes, not an exact material shock crossing. Cadence limitations, uncertain endpoints and direction ambiguity are explicit flags. Birth inside is left-censored; an unfinished run leaves a separately readable suspended right-censored view.

Restart phase	Generation	Exact resumed rows	Exact events
outside	1	41,600	128
active	61	33,920	128
pending	68	33,024	128

The strict reader detects no duplicate canonical event identities in these runs. Event keys are (lineage, unsigned 64-bit particle ID, unsigned 64-bit sequence). Complete encounter state migrates/compacts with the particle. Actual outflow records are emitted before particle deletion.

Per-generation event chunks have bounded capacity and immutable commits. Restart pairs the native checkpoint with a full encounter sidecar and requires the retained history prefix. Checks bind implementation, configuration, time, count, native layout and partition; missing, changed or incompatible data reject explicitly.

This is same-source, same-layout restart evidence. It is not an all-rank atomic transaction or power-loss/storage-fault qualification. MPI builds/singleton checks pass; the local two-rank launcher failed before AthenaK because of runtime compatibility. True distributed migration/restart and target filesystems remain untested.

Appendix F. Measured laptop performance

Uniform flow, 1,024 tracers, t_limit=0.1; two meshes and three modes. Three repetitions per mode/mesh, with execution order rotated. All 18 benchmark runs completed. Times below are whole-process medians, including startup and ordinary outputs.

Mesh	Tracking off	Tracking on	Tracking + trace
64 x 4	0.01163 s	0.04171 s (3.59x)	0.28880 s (24.84x)
256 x 4	0.04174 s	0.12862 s (3.08x)	1.10772 s (26.54x)

Tracking costs about 3.1-3.6x in these tiny jobs. Full observation/transfer traces cost about 25-27x. Durable empty event chunks are still committed every generation; trace mode writes every sampled observation and transfer. These costs must be visible when choosing operational output settings.

Output-cost figures report actual shock diagnostic bytes and file counts, excluding fluid/thermo output. Resident memory was not measured. Short local jobs do not establish sustained production overhead, scaling, MPI/GPU performance or costs on RUSTY/Frontier.

Reproduction evidence and report scope

The initial fresh engineering and physical suites took 13.83 and 28.05 seconds. Their raw evidence occupied about 1.2 GB. The plotted workflow was independently repeated without reuse in a fresh directory: all 21 physical measurement sets and 27 PNGs reproduced exactly, including output cost. The runtime figure retained fresh measurements.

The first plot attempt hit a Matplotlib layout error on an all-zero restart axis. Only the display scale/label coordinates were fixed; exact-equality assertions were retained. A packaging count was corrected from 26 to 27 matching figures because the output-cost plot also reproduced. These were reporting issues, not numerical fixes.

Remaining gates

Human review and acceptable recovery/duplicate/timing/strength thresholds; RNG and frame dependence; sustained encounters and turbulence; true inter-rank operation; threaded/GPU/single precision and cluster budgets. AMR, cooling/source terms, MHD and relativistic physics require separate qualification. Material loops/strings are excluded.

Appendix G. Run the tests and locate the evidence

Run from the development worktree using the existing Release executable. Use new work and output directories each time. The existing ShockIt Python environment contains NumPy, Matplotlib and h5py; the PDF authoring step uses the separate bundled document runtime, with ReportLab and pypdf.

```
cd /Users/hitesh/.codex/worktrees/shock-multiphase/athenak_nyu
MPLCONFIGDIR=/tmp/athenak-mpl \
/Users/hitesh/hitesh/git/shockit/.venv/bin/python \
  tst/shocks-mc/laptop_visuals.py \
  --executable build-release/src/athena \
  --workdir /tmp/athenak-report-run-new \
  --output /tmp/athenak-report-plots-new
open /tmp/athenak-report-plots-new/index.html
```

The existing plot runner generates the raw tests, gallery and plot-only PDF. To produce this explained report from a new packet, run the following separate step:

```
/Users/hitesh/.cache/codex-runtimes/\
codex-primary-runtime/dependencies/python/bin/python3 \
  tst/shocks-mc/explained_pdf.py \
  --packet /tmp/athenak-report-plots-new \
  --plots /tmp/athenak-report-plots-new/all-plots.pdf \
  --output /tmp/athenak-report-explained.pdf
```

That PDF step consumes existing results; it does not rerun the simulation. For a fresh build: cmake -S . -B build-release -DCMAKE_BUILD_TYPE=Release -DAthena_SHOCK_TESTS=ON; then cmake --build build-release -j8.

Source and evidence identity

Core implementation fingerprint:

```
3feb3bcdf77d21296516544e67b5ad75f6ac6ca7bc17c4b48be886ddc0fb7cbc
```

Executable SHA256:

```
0b78478d9b550584051c8a564c562d9bf76628d88f66ddb3ab24aaaa368aff2d
```

Pilot input: inputs/tests/shock_mc.athinput

Test runners: tst/shocks-mc/integration.py, qualification.py, laptop_visuals.py

Recorded packet: evidence/shocks-mc/laptop-visuals/

Parity/persistence review: evidence/shocks-mc/review3/

Schema/support: docs/shocks-mc/schema.md, README.md, human-review.md

Recorded versions: Python 3.14.0, NumPy 2.5.0, Matplotlib 3.11.0, h5py 3.16.0

Scientific sign-off: pending. CPU Serial/double/RK2/uniform Cartesian ideal-gas hydro is the measured local scope. Earlier evidence retains its own source identity.